

量子计算与超导量子芯片简介

刘玉玺

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内容

- 研究背景
- 什么是量子计算
- 怎么造量子计算机
- 基于超导量子芯片的量子计算

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最近的新闻

美国商务部近年来持续发布针对中国芯片产业在半导体、量子计算和人工智能三大领域的限制和禁运措施。

“量子” 计算

近几年多个诺贝尔奖与“量子”的研究相关

2005年，美国科学家罗伊·格劳伯、约翰·霍尔和德国科学家特奥多尔·亨施因为“对光学相干的量子理论的贡献”和对基于激光的精密光谱学发展。[光量子相干](#)

2012年诺贝尔物理奖获得者：Serge Haroche and David J. Wineland，“能够测量和操控单个量子体系的开拓性实验方法”。[单量子操控](#)

2016年，英国科学家大卫·索利斯、邓肯·霍尔丹和迈克尔·科斯特利茨，因在理论上发现了物质的拓扑相变和拓扑相而荣获该奖项。[拓扑量子计算](#)

2022年诺贝尔物理学奖公布获奖者阿兰·阿斯佩、约翰·弗朗西斯·克劳泽、安东·塞林格，表彰他们通过光子纠缠实验，确定贝尔不等式在量子世界中不成立，并开创了量子信息这一学科。[量子信息](#)

2023年诺贝尔化学奖授予美国麻省理工学院教授蒙吉·巴文迪、美国比亚大学教授路易斯·布鲁斯和纳米晶体科技公司科学家阿列克谢·伊基莫夫，表彰他们在“发现和合成量子点”方面的贡献。[量子点用量子力学](#)

奇谈怪论

罗杰·彭罗斯：黑洞理论的贡献获得 2020年的诺贝尔物理奖，他提出意识的本质是量子计算，用量子力学解释灵魂。



图片来源：视觉中国 www.vcg.com

第二次“量子”革命



第一次量子革命

主要是科学层面（在19和20世纪之交）：到20世纪末导致了现代社会的很多核心技术，如：激光、计算机芯片工业（信息时代）、太阳能、复印机等等。用量子分离性。

第二次量子革命将掀起技术创新的巨大浪潮

主要是技术层面（20世纪末）：引领21世纪的关键物理技术，如上世纪80年代初提出的量子信息处理。用量子相干性。

量子



计算机

计算机发展简史



算盘

中国算盘最早追溯到中国汉朝（**190年**），日本安土桃山时代（**1575年**）。



机械计算器：1930年前

注：现代计算机的真正起源来自英国数学教授**Charles Babbage**。他构想和设计一台完全可编程计算机的第一人。



电子计算器：1930年-1950年

ENIAC：电子数字积分与计算器，1946年。

冯诺伊曼加入研制**ENIAC**，提出储存程序计算机的构想。

注：1936年艾伦·图灵 提出通用计算机的数学模型（图灵机），奠定了计算机的理论基础。

计算机的核心构件



晶体管



芯片

数字计算机：1950年—



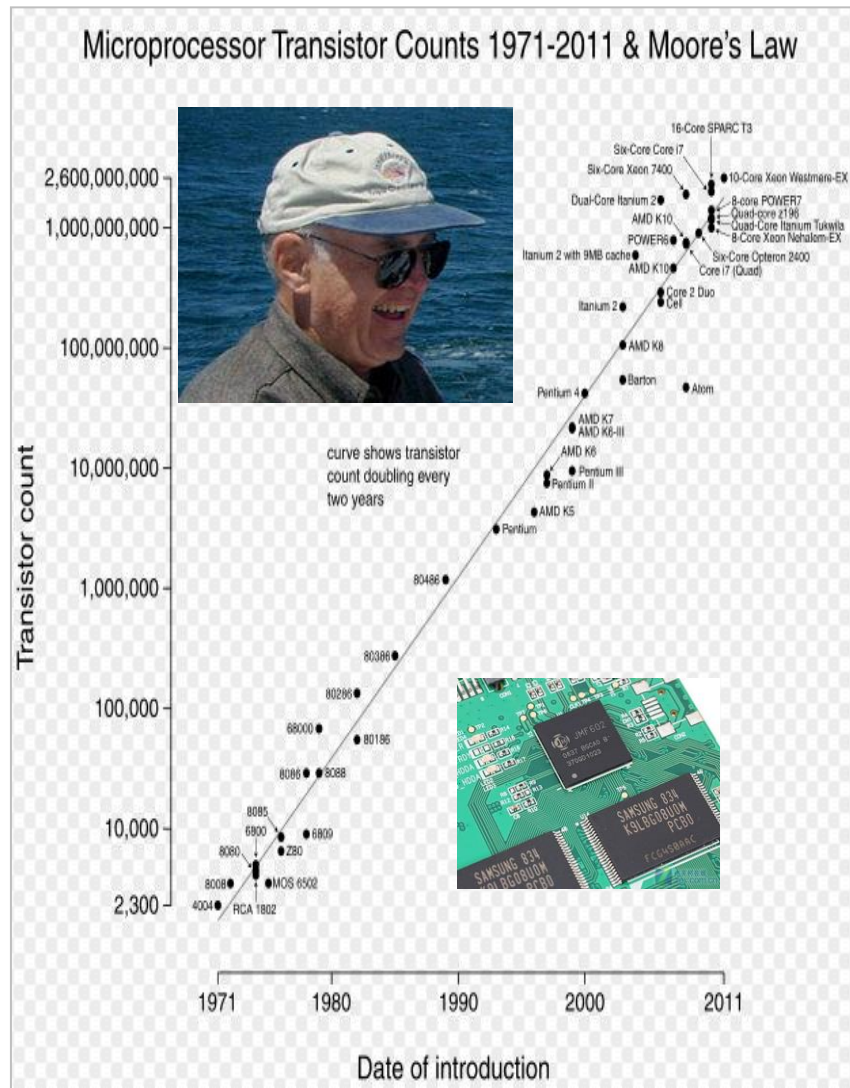
晶体管的出现：威廉·肖克利、约翰·巴丁和沃尔特·布拉滕，发明了晶体管，被誉为**晶体管之父**，1956年获诺贝尔奖。巴丁1956年和1972年两次获诺贝尔物理学奖。

摩尔定律

摩尔定律（1965年）：单位芯片信息
元器件数目18（或24）个月增加一倍，
性能提升一倍。

ITRS（国际半导体技术发展路线图）：
2016年集成电路主流工艺线宽将为22纳米，2022年将到10纳米。**实际情况：**台积电2018年7纳米，台积电、三星2023年推出3纳米。

国人头发的粗细：85000纳米



戈登摩尔是英特尔（Intel）创始人之一。

晶圓代工 2 奈米以下比較

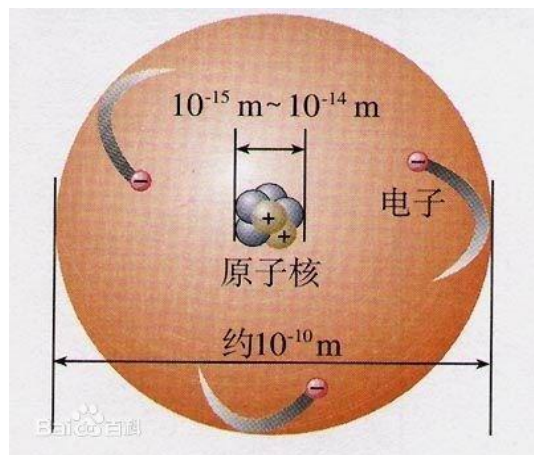
TechNews
科技新報

公司	台積電		英特爾		三星	
製程名稱	N2	A14	20A	18A	SF2	SF 1.4
製程 (nm)	2	1.4	2	1.8	2	1.4
製程結構	GAAFET (Nanosheet)	GAAFET (Nanosheet)	RibbonFET	RibbonFET	GAAFET (Nanosheet)	GAAFET (Nanosheet)
量產時間	2025 年	2027 年	2024 年 H1	2024 年 H2	2025 年	2027 年
備註	* 時間點依路線圖				3 奈米製程：GAA MBCFET	

製表：科技新報

业界普遍认为28nm是成熟制程与先进制程的分界线，28nm及以上的制程工艺被称为成熟制程，28nm以下的制程工艺被称为先进制程。先进制程技术不断突破极限，目前已发展到了2nm，**当芯片制程接近1nm时，就进入量子物理的世界，现有的工艺制程会受到量子效应的极大影响，很难再进一步。**

信息器件的物理极限



原子线度：1埃 = 0.1纳米

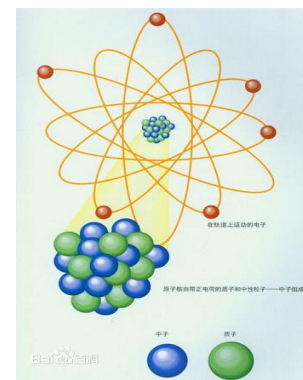
1纳米相当于排列10个原子。

信息元器件的尺寸趋于**经典物理极限**，欧姆定律不再成立，**量子现象显现**。

以硅为基础的信息器件发展所遵循的摩尔定律不再适用。需要研究纳尺度器件的性能。



怎么办？利用“量子”做文章，采用不同的计算模式。



量子计算机概念的出现和通用量子计算机的提出

“杞人忧天”的物理学家

摩尔定律会永远成立吗？如何解决复杂结构物质的计算机仿真？

计算机的量子力学模型



保罗·贝尼奥夫主要从事核化学相关的工作。1980年，他将图灵机中的每个组成部分和操作在他构造出的基于量子系统的计算机模型中找到了对应，构造出无能量耗散的计算模型，本质还是经典的。

The Computer as a Physical System: A Microscopic Quantum Mechanical Hamiltonian Model of Computers as Represented by Turing Machines

Paul Benioff^{1,2}

Journal of Statistical Physics, Vol. 22, No. 5, 1980

Received June 11, 1979; revised August 9, 1979

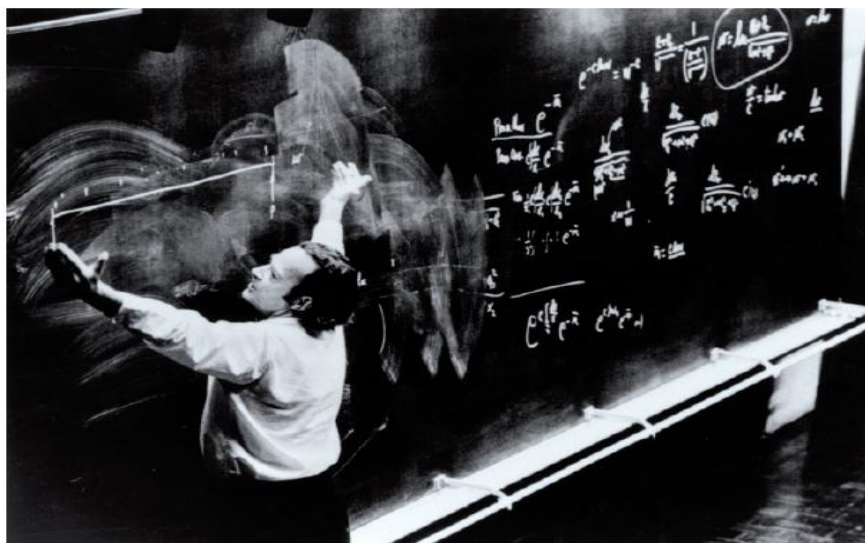
法国国家科学研究院、美国阿贡国家实验室

量子计算概念的出现：量子系统仿真大系统

4. QUANTUM COMPUTERS—UNIVERSAL QUANTUM SIMULATORS

Richard P. Feynman, *Int. J. Theor. Phys.* **21**, 467 (1982).

The first branch, one you might call a side-remark, is, Can you do it with a new kind of computer—a quantum computer? (I'll come back to the other branch in a moment.) Now it turns out, as far as I can tell, that you can simulate this with a quantum system, with quantum computer elements.



费曼构想的“量子模拟机”具备了充分的量子力学特性，可以模拟经典图灵机无法完美模拟的量子物理系统，但并没有进一步考虑赋予其通用性。

在量子电动力学和粒子物理学方面的贡献奖得1965年诺贝尔物理学

通用量子计算的提出



戴维·多伊齐 (David Deutsch) 在牛津沃尔夫森学院获理论物理学博士学位，论文研究弯曲时空的量子场论。

1985 年，他发表了影响深远的论文，讨论了量子图灵机的若干特性，如量子随机性，量子关联，量子并行性，量子算法优势，并非常富有远见地指出了量子计算复杂度理论的研究意义，这些概念都极大地指导了后来量子计算科学的研究。

Proc. R. Soc. Lond. A **400**, 97–117 (1985)

Printed in Great Britain

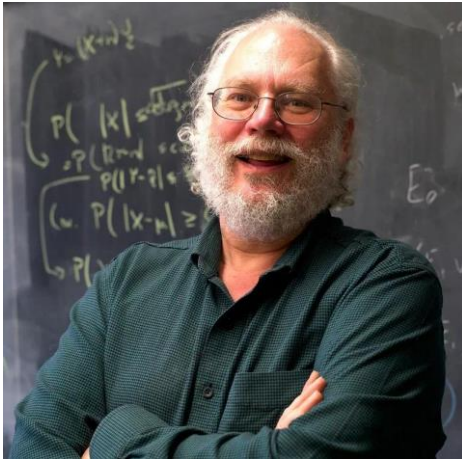
Quantum theory, the Church–Turing principle and the universal quantum computer

BY D. DEUTSCH

Department of Astrophysics, South Parks Road, Oxford OX1 3RQ, U.K.

(Communicated by R. Penrose, F.R.S. – Received 13 July 1984)

量子计算机能干什么？



彼得·秀尔（Peter Shor）在贝尔实验室于1994年开发出了大名鼎鼎的Shor算法。

展示了量子计算可以在质因数分解问题上实现比经典计算机指数级的加速，让人们相信量子计算机可以在模拟量子物理系统之外得到广泛应用的算法，引发了物理学界之外广泛的讨论，开启了量子计算研究的热潮。

Algorithms for Quantum Computation: Discrete Logarithms and Factoring

Peter W. Shor
AT&T Bell Labs
Room 2D-149
600 Mountain Ave.
Murray Hill, NJ 07974, USA

量子计算机
几秒钟就能
把目前最快的
超算不能
分解的一个
大数分解。

P. Shor, in Proc. 35th Annual Symposium on Foundations of Computer Science. IEEE Press, Nov. **1994**.

Shor算法有什么用？

优点：要分解整数 N ，Shor算法需要时间是 $\log N$ 的某个多项式，比传统已知最快的因子分解算法快了指数倍。

重要性：针对公开密钥加密和电子商业中的RSA（Ron Rivest, Adi Shamir, Leonard Adleman）算法最流行的攻击一般是基于大数分解。

目前密钥长度2048比特以上的加密是绝对安全的。

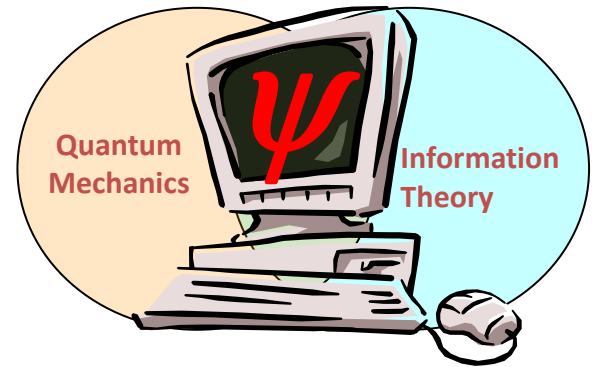
近20多年的发展，发现量子计算还能用于：

目标搜索、新材料合成、生物制药的研发、气象预测、军事科学、人工智能等等。

为什么要研发量子计算

理想：大规模复杂运算的需要，量子论的深化。

现实：信息器件的线度达到物理极限，器件在原有理论上不能工作。**必须发展新的运算方式。**



量子计算机的出现是信息技术发展的必然结果、也是解决复杂科学问题的需求、量子理论的进一步深化。

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量子是什么？为什么这么“淘气”？

“量子”：是一些不能连续变化的物理量（如能量）中最小不可分割单位的统称，这个最小单位就叫做“量子”，是一抽象概念，前面加定语才能得到更具体的理解。

与宏观物体不一样，描述“量子”的运动，需要与牛顿力学不一样的量子力学。

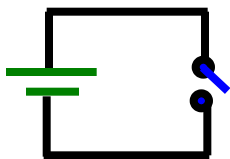
“量子”最“淘气”之处是其所处的状态不具有确定性。

什么是量子计算？

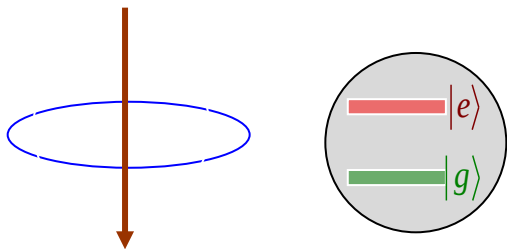
量子计算是基于量子力学原理（比如，态叠加和纠缠）对信息执行编码、逻辑运算、存储及处理。

简而言之：量子物理和计算机科学的结合体。

量子信息处理的基本单元：量子比特



经典比特只能是0态或1态。



量子双态系统 $\Leftrightarrow$ 量子比特

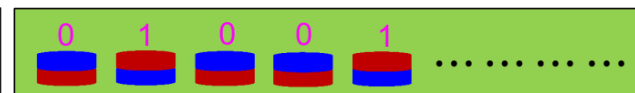
核心：是 $|0\rangle$ 态和 $|1\rangle$ 态的量子相干叠加。

经典比特

1比特



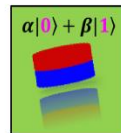
N 比特



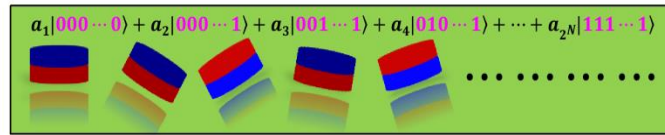
状态、操控、工作方式不同

量子比特

1比特



N 比特



量子比特

量子比特：英文为**Qubit**，是**Quantum bit** 的缩写。它是一个双态（二能级）的量子系统。

双态量子系统：有两个独立可分的量子状态 $|0\rangle$ 和 $|1\rangle$ 。

量子比特的核心：能够处在状态的 $|0\rangle$ 和 $|1\rangle$ 的叠加：

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

经典比特只有0和1两个状态；量子比特由于相干叠加从而有无穷多个状态。

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如何实现量子计算



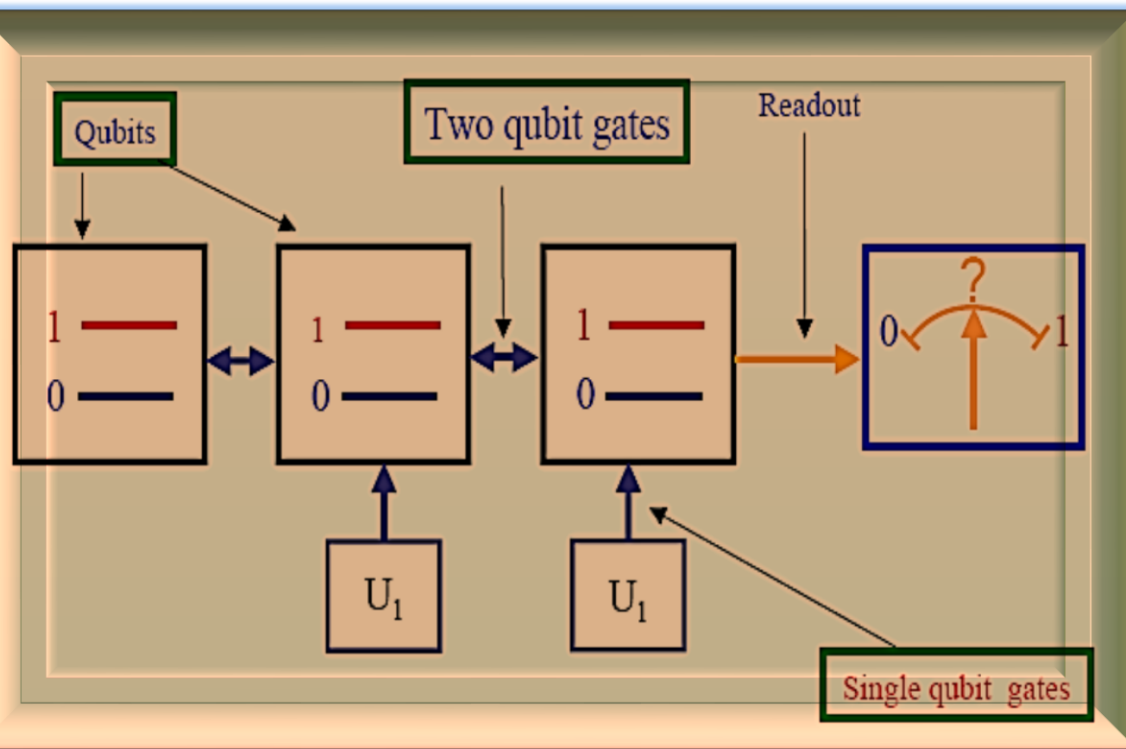
量子计算包括：硬件（量子芯片）、编码、和算法（软件）三部分

主要在物理上寻求克服量子消相干的方法，制备好的量子比特。

量子计算基本蓝图

DiVincenzo 规则

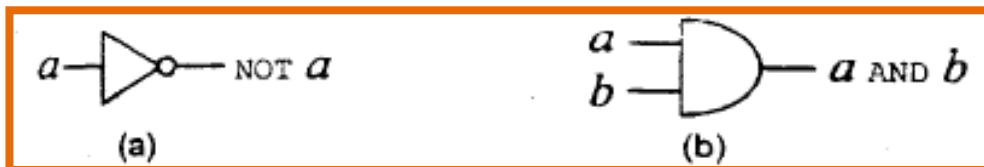
- ◆ 长的相干时间
- ◆ 读出或测量
- ◆ 初始化
- ◆ 通用的逻辑门：由可控的哈密顿实现
- ◆ 可拓展的量子比特



找量子比特=》根据上述规则即可造计算机。

量子逻辑门

经典的非门、与门等



单比特量子逻辑门

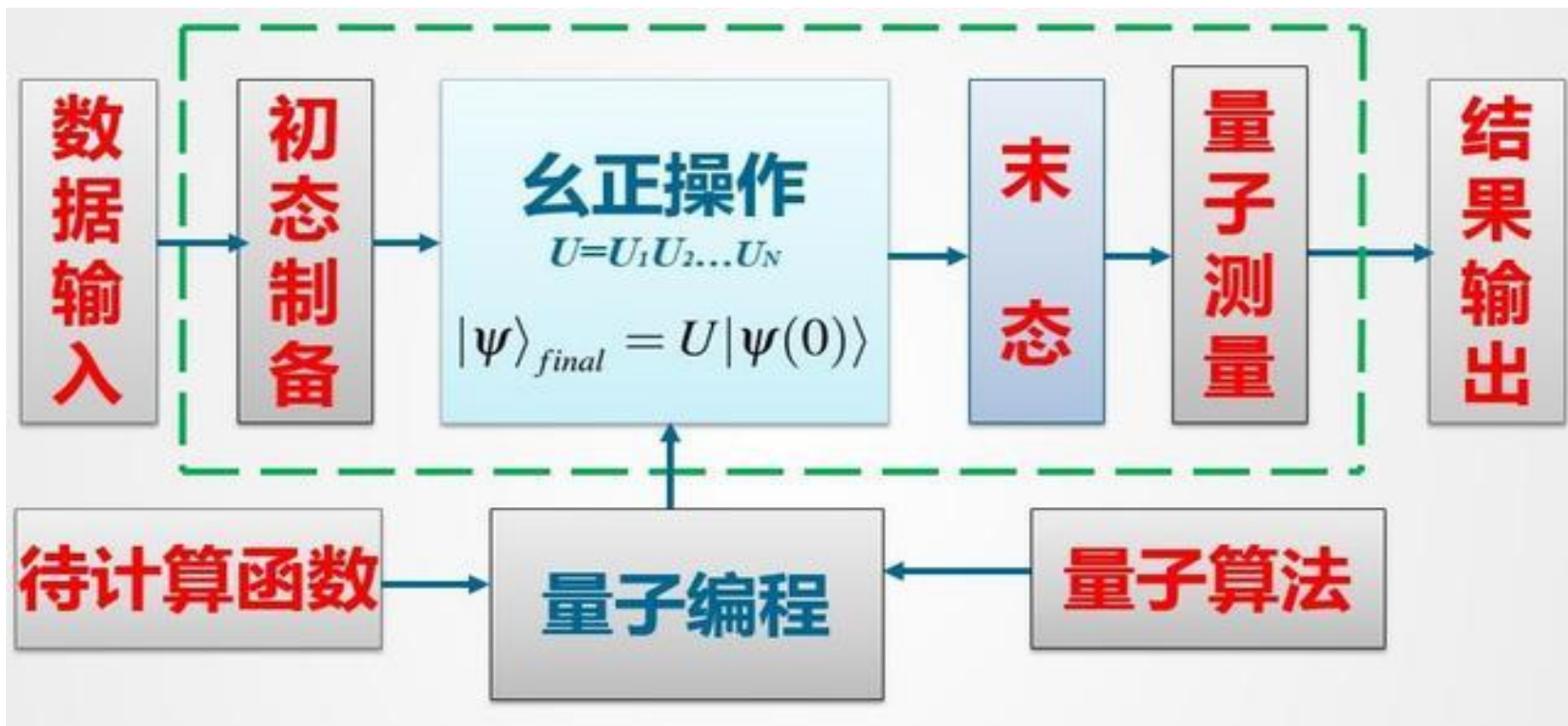
- 可用 2×2 矩阵来表示。
- 此矩阵必须是酉阵或说么正矩阵“酉”为唯一的限制。
- 任何 2×2 酉矩阵都构成单量子门。
- X、Y、Z、Hadamard门等。

双比特量子逻辑门

CNOT门、SWAP门等

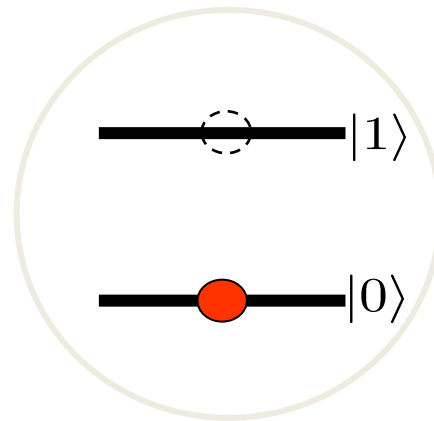
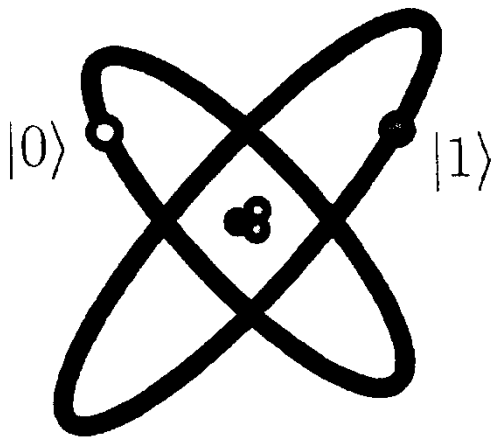
名称	符号	酉阵
Hadamard		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Pauli-X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Phase		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

量子计算的功能图



上图来源于郭光灿院士报告

量子比特的物理实现



$$\alpha|0\rangle + \beta|1\rangle$$

实现形式:

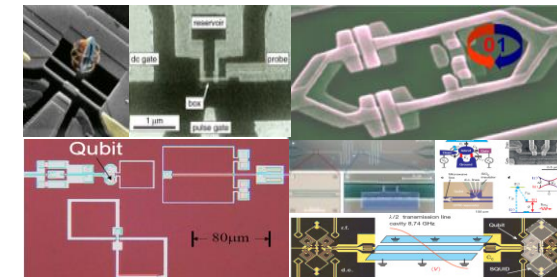
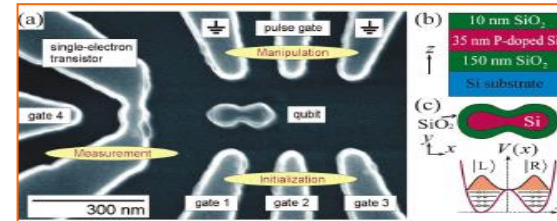
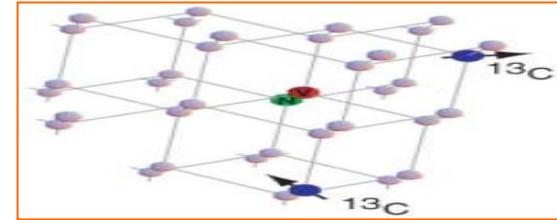
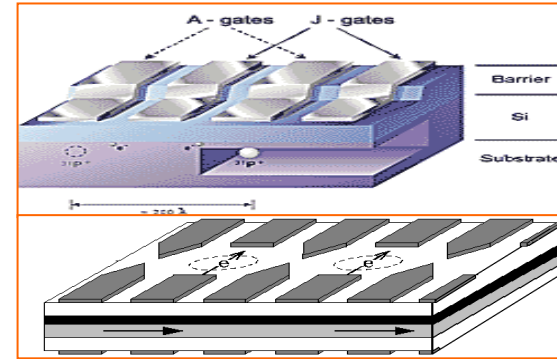
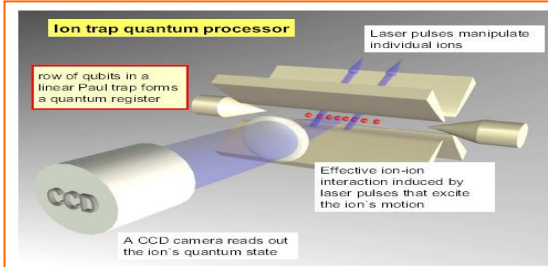
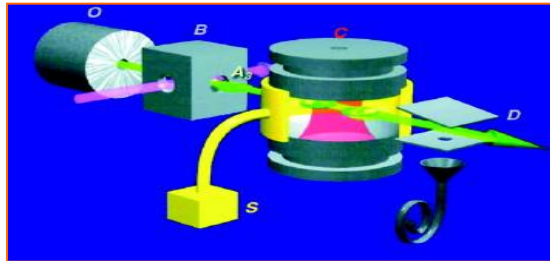
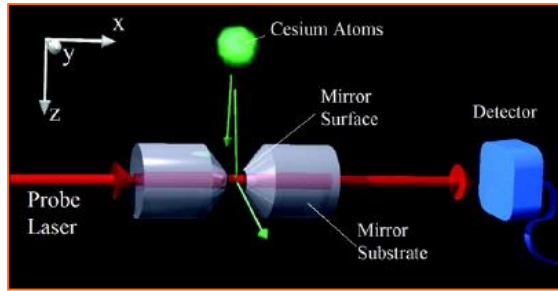
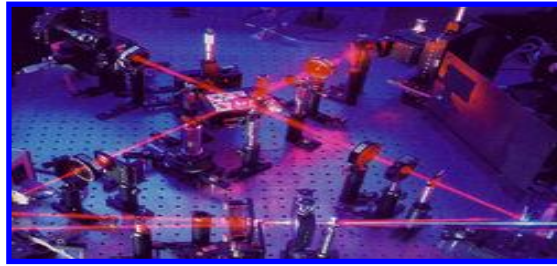
光子的不同的偏振态； 原子； 核子和电子自旋；
电子绕单原子核的运动状态； 超导体中的电流、电荷等。

◆经过近30年的研究，量子计算机实现的基本原理没有任何障碍！

◆关键在硬件，找好的量子相干器件，即量子芯片，然后发展集成量子芯片技术。

各种量子双态系统

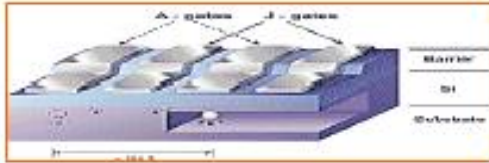
天然原子光子等系统



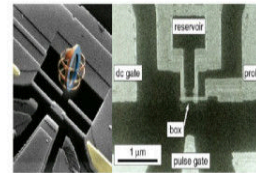
人工固态结构系统

固体量子相干器件的优点

半导体系统

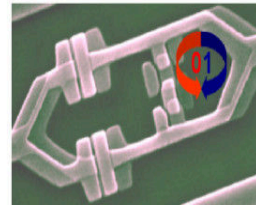


超导量子电路和人工原子



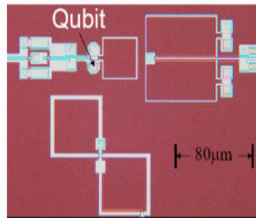
NEC Tsukuba (1999)
Chalmers/Yale
Saclay

$$E_J = E_C \text{ or } 0.1 E_C$$



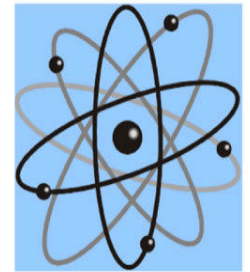
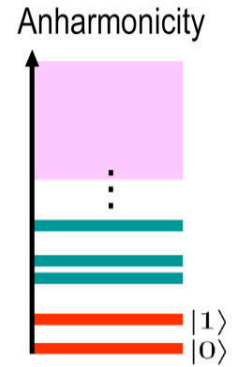
Delft
MIT
NTT
NEC
Berkeley
Munich

$$E_J = 40\text{--}100 E_C$$



$$E_J = 10,000 E_C$$

UCSB
Kansas
NanJing Univ.
NIST Boulder
Maryland



易控制、集成性好、得到先进纳米制备技术的支持，可以基于半导体微加工技术加以实现，而超导量子电路走到最前面。

内容

- 研究背景
- 什么是量子计算
- 怎么造量子计算机
- **基于超导量子芯片的量子计算**



教授

John Clarke, born 1942 in Cambridge, UK. PhD 1968 from University of Cambridge, UK. Professor at University of California, Berkeley, USA.



博士后

Michel H. Devoret, born 1953 in Paris, France. PhD 1982 from Paris-Sud University, France. Professor at Yale University, New Haven, CT and University of California, Santa Barbara, USA.



博士生

John M. Martinis, born 1958. PhD 1987 from University of California, Berkeley, USA. Professor at University of California, Santa Barbara, USA.

The Nobel Prize in Physics 2025 is awarded to John Clarke, Michel H. Devoret and John M. Martinis

A major question in physics is the maximum size of a system that can demonstrate quantum mechanical effects. This year's Nobel Prize Laureates conducted experiments with an electrical circuit in which they demonstrated both quantum mechanical tunnelling and quantised energy levels in a system big enough to be held in the hand.

PROSPECTS IN ULTRALOW TEMPERATURE PHYSICS

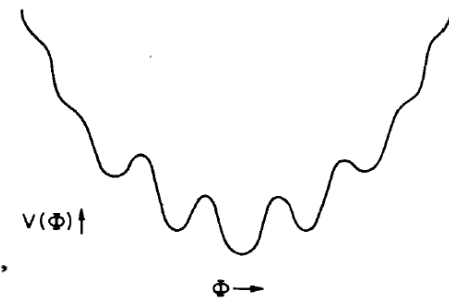
$$U(\Phi) = \frac{1}{2} \frac{(\Phi - \Phi_K)^2}{L} - i_c \frac{\Phi_0}{2\pi} \cos \frac{2\pi\Phi}{\Phi_0}$$

A.J. Leggett

School of Mathematical and Physical Sciences, University of Sussex, Brighton BN1 9QH, Sussex, U.K.

Résumé.- Comme avenir de la recherche sur les très basses températures, on peut prévoir par exemple, la découverte de nouvelles phases exotiques, l'observation éventuelle du "percement macroscopique quantique" dans le vide, et l'amplification des interactions très faibles de la physique des hautes énergies.

Abstract.- Why the continuing search for ultralow temperatures? Possible motives include the prospect of exotic new phases, of observing macroscopic vacuum quantum tunnelling and of amplifying the ultra-weak interactions postulated in particle physics.



富勒烯大分子，磁化强度，机械振动

PHYSICAL REVIEW
LETTERS

VOLUME 54

4 MARCH 1985

NUMBER 9

Quantum Mechanics versus Macroscopic Realism: Is the Flux There when Nobody Looks?

A. J. Leggett

Department of Physics,^(a) University of Illinois at Urbana-Champaign, Urbana, Illinois 61801, and Department of Physics, Harvard University, Cambridge, Massachusetts 02138

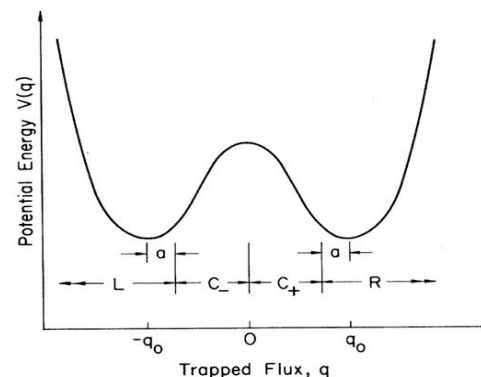
and

Anupam Garg

University of Illinois at Urbana-Champaign, Urbana, Illinois 61801

(Received 19 November 1984)

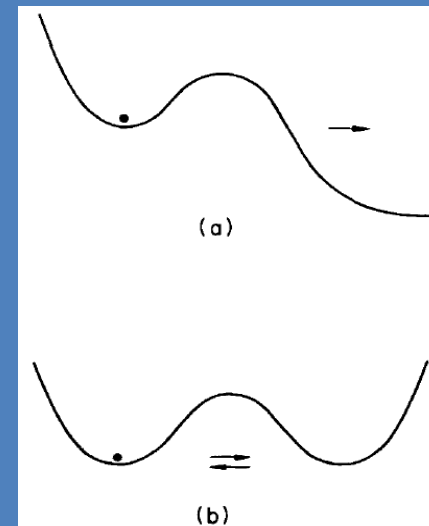
It is shown that, in the context of an idealized "macroscopic quantum coherence" experiment, the predictions of quantum mechanics are incompatible with the conjunction of two general assumptions which are designated "macroscopic realism" and "noninvasive measurability at the macroscopic level." The conditions under which quantum mechanics can be tested against these assumptions in a realistic experiment are discussed.



CONTEMP. PHYS., 1984, VOL. 25, NO. 6, 583-598

Schrödinger's Cat
and her Laboratory Cousins

A. J. Leggett, Department of Physics, University of Illinois at Urbana-Champaign, Illinois 61801, USA



2003年诺贝尔物理奖

Prize motivation: "for pioneering contributions to the theory of superconductors and superfluids"

诺奖主要工作

VOLUME 53, NUMBER 13

PHYSICAL REVIEW LETTERS

24 SEPTEMBER 1984

Resonant Activation from the Zero-Voltage State of a Current-Biased Josephson Junction

Michel H. Devoret,^(a) John M. Martinis, Daniel Esteve,^(a) and John Clarke

Department of Physics, University of California, Berkeley, California 94720, and Materials and Molecular Research Division, Lawrence Berkeley Laboratory, Berkeley, California 94720

(Received 26 July 1984)

The lifetime τ of the zero voltage limit has been measured. The microwave frequency is chosen to be resonantly activated. The results are well explained by the theory of macroscopic quantum tunneling.

VOLUME 55, NUMBER 18

PHYSICAL REVIEW LETTERS

28 OCTOBER 1985

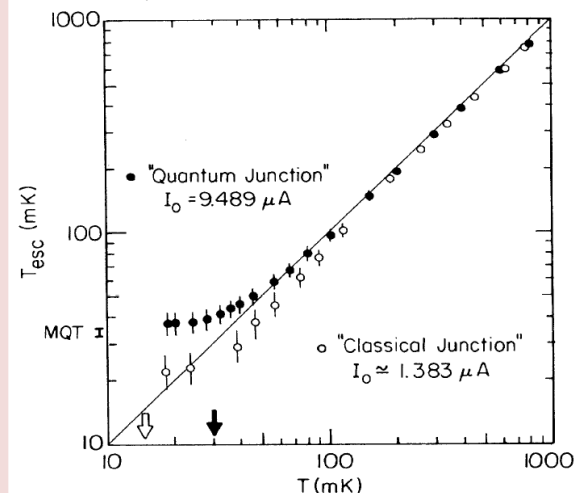
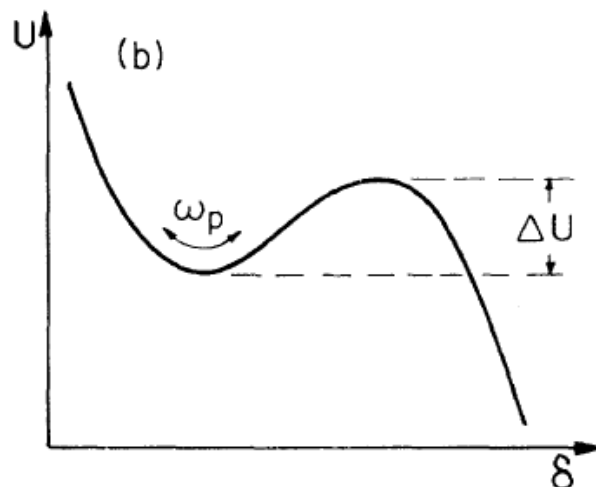
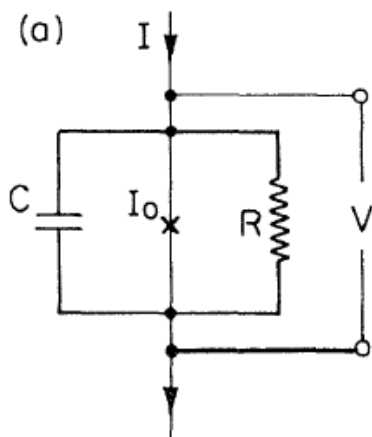
Measurements of Macroscopic Quantum Tunneling out of the Zero-Voltage State of a Current-Biased Josephson Junction

Michel H. Devoret,^(a) John M. Martinis, and John Clarke

Department of Physics, University of California, Berkeley, California 94720, and Materials and Molecular Research Division, Lawrence Berkeley Laboratory, Berkeley, California 94720

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诺奖主要工作

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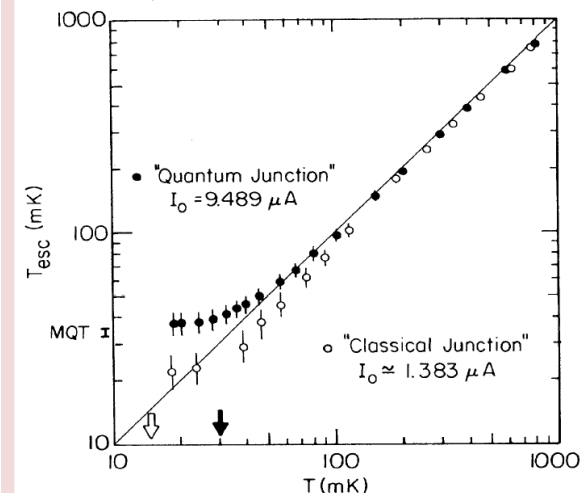
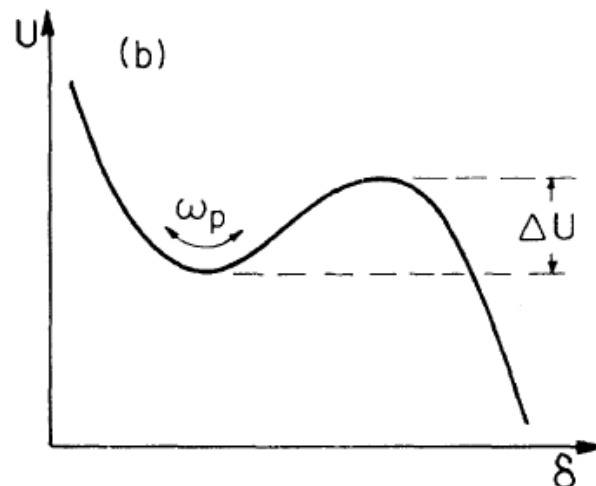
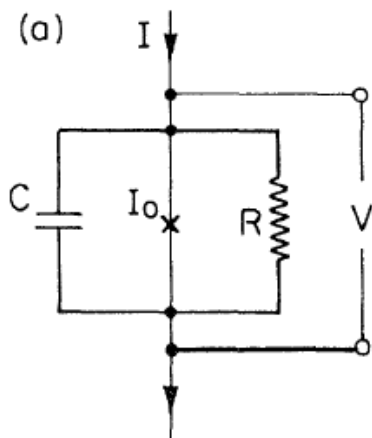
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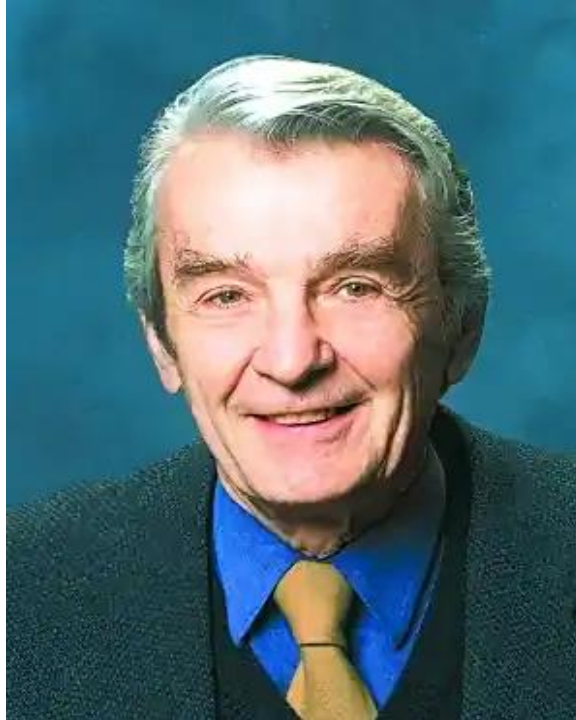


Brian David Josephson
Cambridge University

Nobel Prize - Physics 1973

for his theoretical predictions of the properties of a super-current through a tunnel barrier, in particular those phenomena which are generally known as the Josephson effects

理论提出库珀对的量子隧穿效应



Ivar Giaever (5 April, 1929-20 June, 2025)
General Electric Company,
Schenectady, NY

Nobel Prize - Physics 1973

for their experimental discoveries regarding tunneling phenomena in semiconductors and superconductors, respectively

证实了半导体和超导体中电子的量子隧穿现象，超导能隙的存在



Leo Esaki (江崎玲于奈) born 1925 in Osaka Japan
IBM Thomas J. Watson Research Center, Yorktown Heights, NY

Nobel Prize - Physics 1973

for their experimental discoveries regarding tunneling phenomena in semiconductors and superconductors, respectively

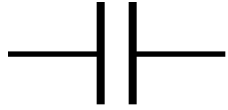
提出了半导体器件中电子的量子隧穿现象



2018年5月浙江大学



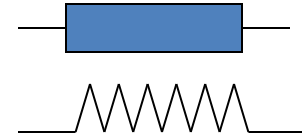
基本元件



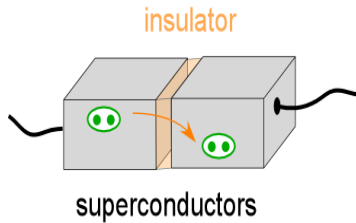
Capacitance: C



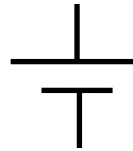
Inductance L



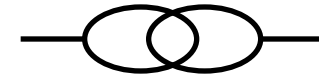
Resistor R



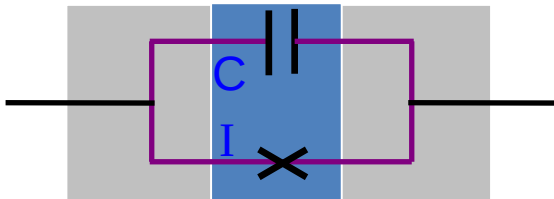
Josephson junction



Voltage: V



Current source I

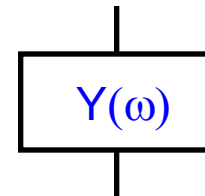


Nb $\Rightarrow$ 1.50 meV; 9.25 K
Al $\Rightarrow$ 0.18 meV; 1.18 K

eV $\Rightarrow$ 1.16×10^4 K $\Rightarrow$ 2.4×10^5 GHz

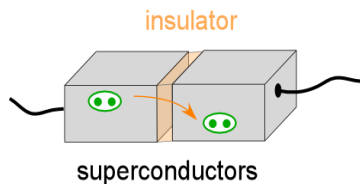


Connected to the ground



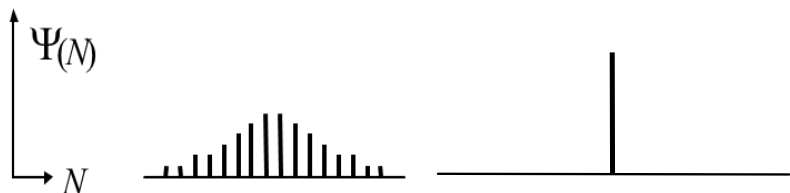
Admittance
Inverse of the impedance $Z(\omega)$

超导量子电路



1999年第一个实验，相干时间2纳秒

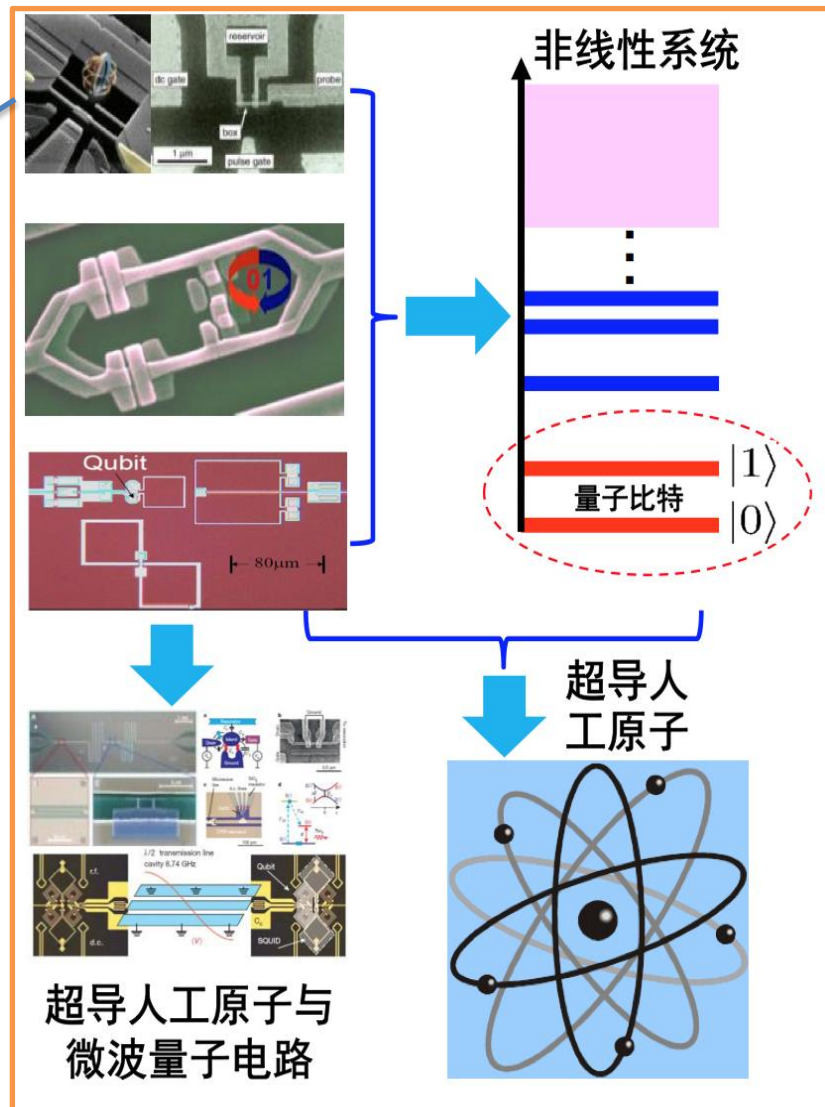
基于约瑟夫森结和纳尺度电子学元件的电路。



位相确定

电荷确定

最初三类：位相，磁通，电荷量子比特



约瑟夫森结的性质决定了超导量子比特工作在微波区域

稀释制冷机

1951年H. London提出可以用超流 ^4He 稀释 ^3He 的方法制冷的理论。到1965年P. Das等人根据这一理论制成了 ^3He - ^4He 稀释制冷机，目前已达到1.8 mK的低温。

摄氏温度 $\leftrightarrow$ 开氏温度

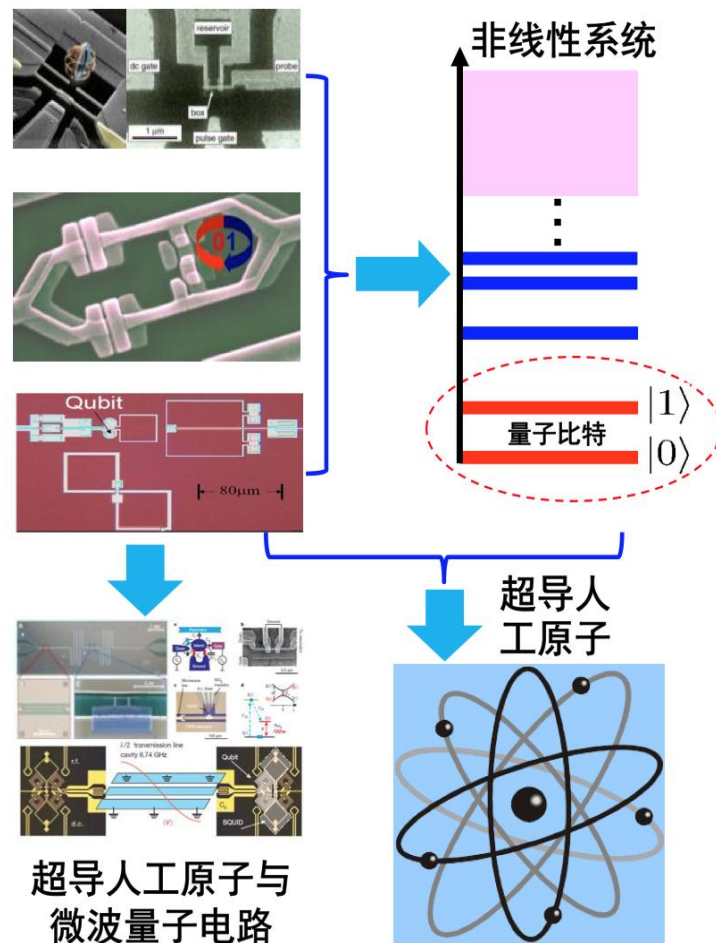
$$T(^{\circ}\text{C}) + 273 = T(\text{K})$$



超导量子电路2007年前的研究

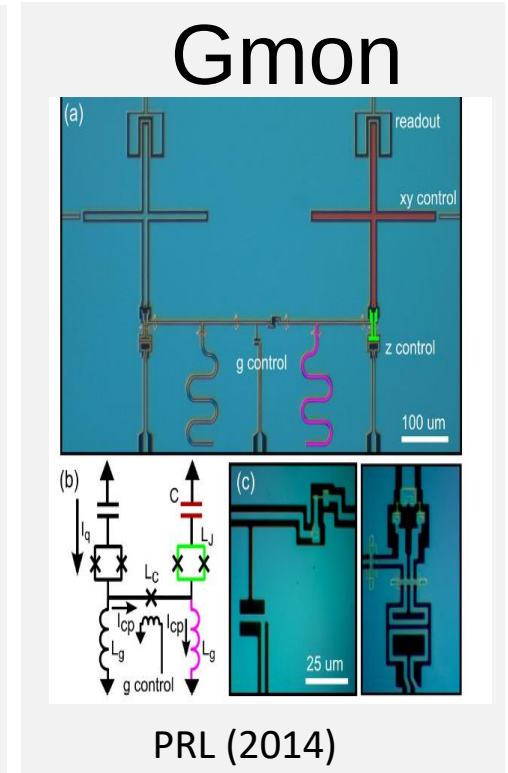
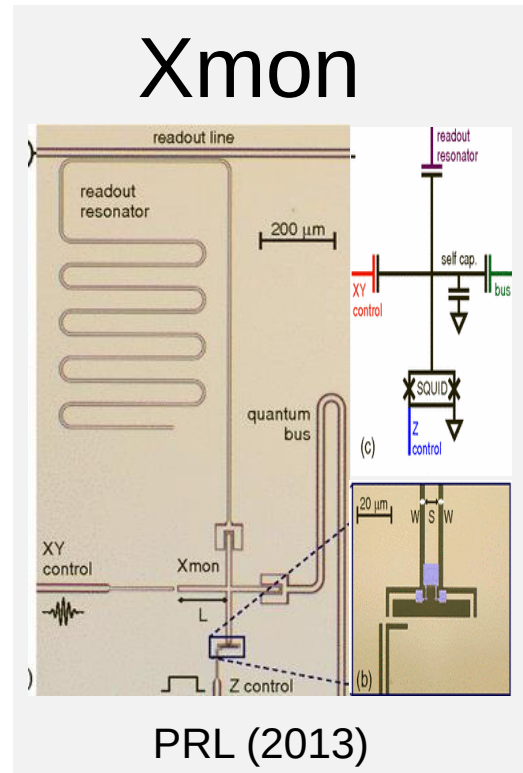
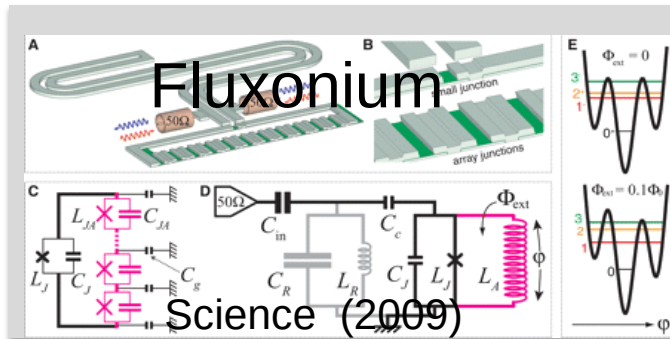
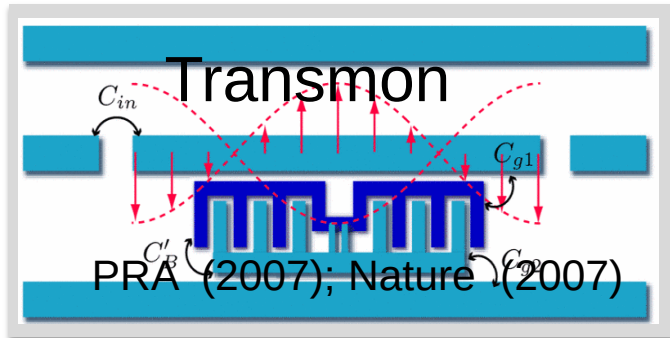
- ◆ 超导人工原子的相干性质：约1999年开始（NEC, Tsai）
- ◆ 超导电路量子电动力学：实验2004年（Yale 和 Delft）
- ◆ 量子态的测量
- ◆ 可控耦合与电路的扩展途径
- ◆ 普适逻辑门与算法演示

超导量子人工原子



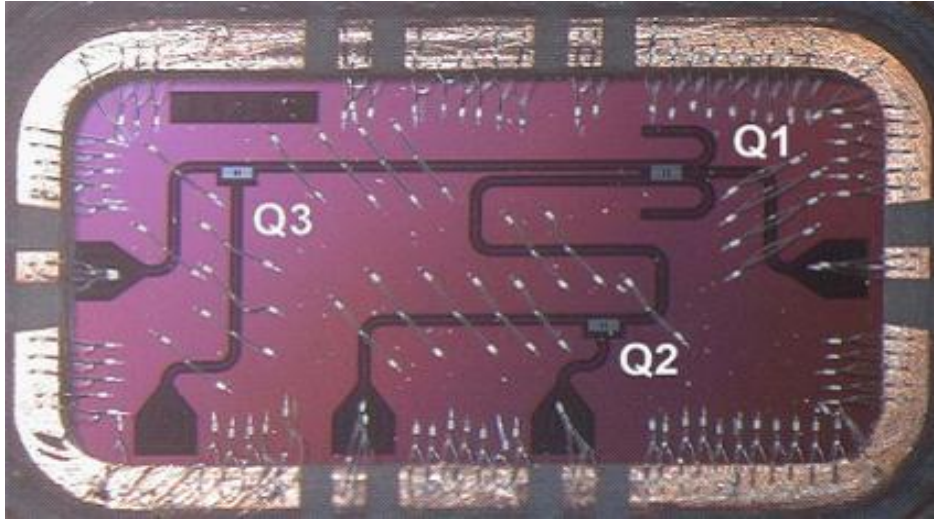
新一代超导量子电路

解决问题：长相干、易互联、快操控、可扩展



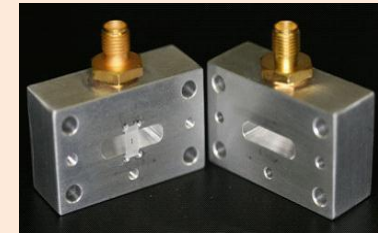
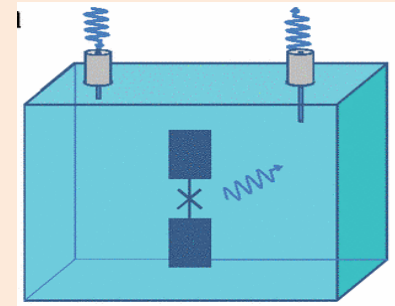
2007年后出现的新型超导量子比特电路

新一代超导量子电路



2011年相干时间达到100微秒量级

三维空间

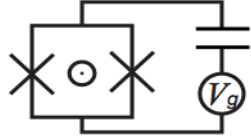
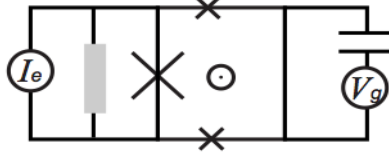
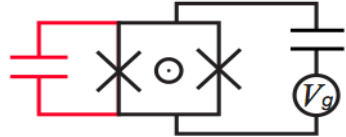
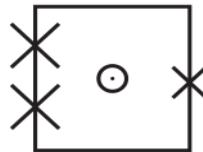
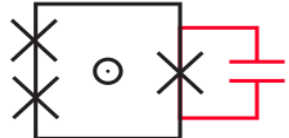
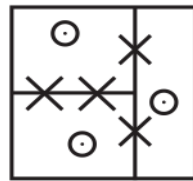
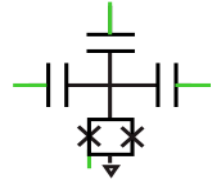
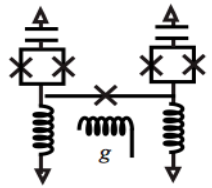


PRL **107**, 240501 (2011)

PRB **86**, 100506 (2012)

特点：长相干时间，电路在空间构型上具有一定的突破

新型超导量子比特性能的提高

Circuit	Split Cooper pair box 	Quantronium 	Transmon 
Improvements	Tunable Josephson energy external flux	Phase qubit readout	Insensitive to charge noise
Circuit	3-junction flux qubit 	Low decoherence 	Fluxonium 
Improvements	Minimize flux noise	Minimize charge noise	Minimize charge noise
Circuit	Tunable Gap 	Xmon 	Gmon 
Improvements	Tunability	Connectivity	Tunable coupling

超导人工原子和微波光子学

Xiu Gu, Anton Frisk Kockum, Adam Miranowicz, Yu-xi Liu, Franco Nori,

Microwave photonics with superconducting quantum circuits, 101 pages

Phys. Rep. 718-719, 1 (2017)

Physics Reports 718–719 (2017) 1–102



Contents lists available at ScienceDirect

Physics Reports

journal homepage: www.elsevier.com/locate/physrep



Microwave photonics with superconducting quantum circuits

Xiu Gu^{a,b,1}, Anton Frisk Kockum^{b,1}, Adam Miranowicz^{b,c}, Yu-xi Liu^{a,b,d,*}, Franco Nori^{b,e,*}

^a Institute of Microelectronics, Tsinghua University, Beijing 100084, China

^b Center for Emergent Matter Science, RIKEN, Saitama 351-0198, Japan

^c Faculty of Physics, Adam Mickiewicz University, 61-614 Poznań, Poland

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^e Physics Department, The University of Michigan, Ann Arbor, MI 48109-1040, USA



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Microwave photonics

ABSTRACT

In the past 20 years, impressive progress has been made both experimentally and theoretically in superconducting quantum circuits, which provide a platform for manipulating microwave photons. This emerging field of superconducting quantum microwave circuits has been driven by many new interesting phenomena in microwave photonics and quantum information processing. For instance, the interaction between superconducting quantum circuits and single microwave photons can reach the regimes of strong, ultra-strong, and even deep-strong coupling. Many higher-order effects, unusual and less familiar in traditional cavity quantum electrodynamics with natural atoms, have been experimentally observed, e.g., giant Kerr effects, multi-photon processes, and single-atom induced bistability of microwave photons. These developments may lead to improved understanding of the counterintuitive properties of quantum mechanics, and speed up applications ranging from microwave photonics to superconducting quantum information processing. In this article, we review experimental and theoretical progress in microwave photonics with superconducting quantum circuits. We hope that this global review can provide a useful roadmap for this rapidly developing field.

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2.2.3. Superconducting phase-qubit circuits	7
2.3. Various extensions of SQCs.....	7
2.4. Selection rules for microwave-induced transitions.....	9
2.5. Three-level superconducting artificial atoms	10
2.6. Readout of qubit states	10

* Corresponding authors at: Institute of Microelectronics, Tsinghua University, Beijing 100084, China (Y.-x. Liu); RIKEN, Saitama 351-0198, Japan (F. Nori).

E-mail addresses: yuxiliu@mails.tsinghua.edu.cn (Y.-x. Liu), fnori@riken.jp (F. Nori).

¹ These authors contributed equally to this work.

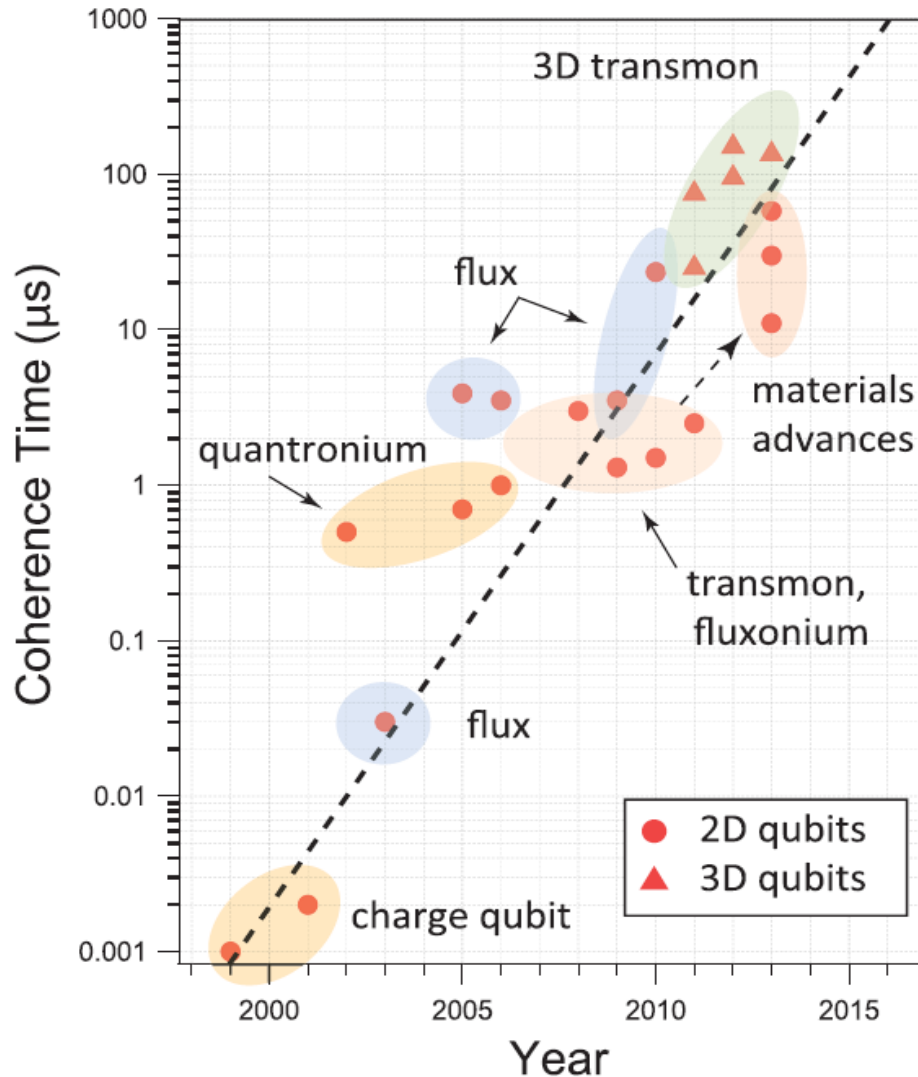
超导量子比特电路相干路线图

可能实现信息的量子处理！

科学家们10多年的努力，
相干时间从纳秒到百微秒，
5 万倍的增加！

寿命达1毫秒级，Yale 大学
Nature 508, 369 (2014)。

目前相干时间达500多微秒



超导量子计算器件的关键问题

DiVincenzo Criteria

可扩展的量子比特 （实验数目每天在更新）

通用逻辑门 （2007年，小型算法的验算实现）

初始化 （没有遇困难）

读出或测量 （没有遇困难，但一直在优化）

长相干时间 （1999–2014）

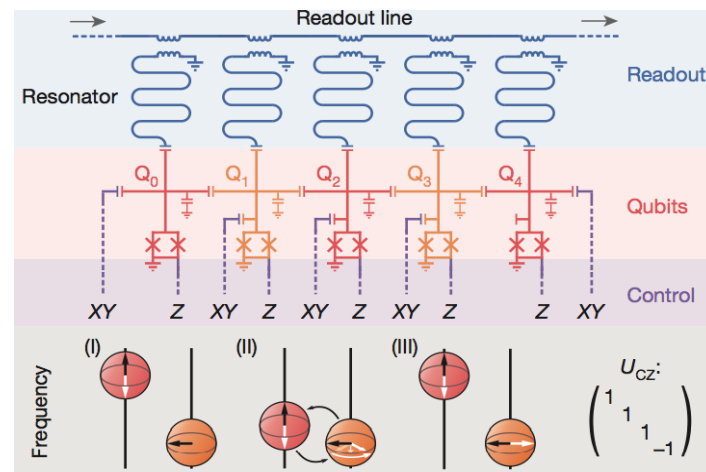
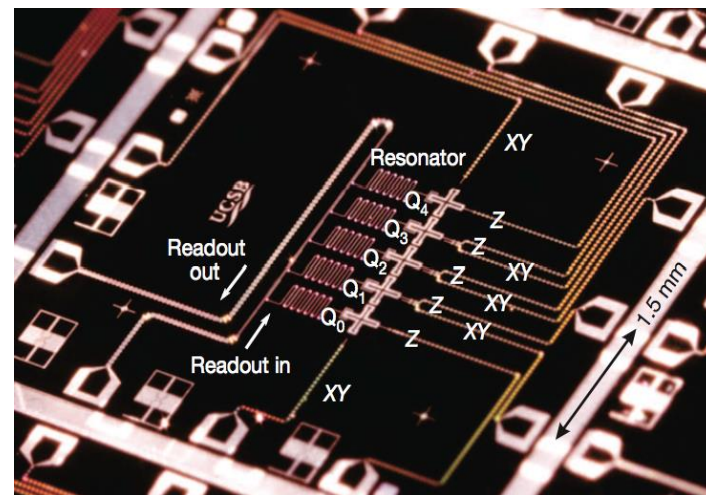
超导量子计算机的实现原则上没有任何障碍！

容错超导量子计算芯片

人们认为 Surface Code 是一种较好的容错编码方案。**优点：**容错阈值大致为每步逻辑操作需要99%的保真度。**缺点：**需要的物理比特太多。

Martinis组实验演示了

- 单比特逻辑门保真度为99.92%
- 双比特逻辑门保真度为99.4%
- 演示了5量子比特的GHZ态
- 具有了容错和错误修正功能



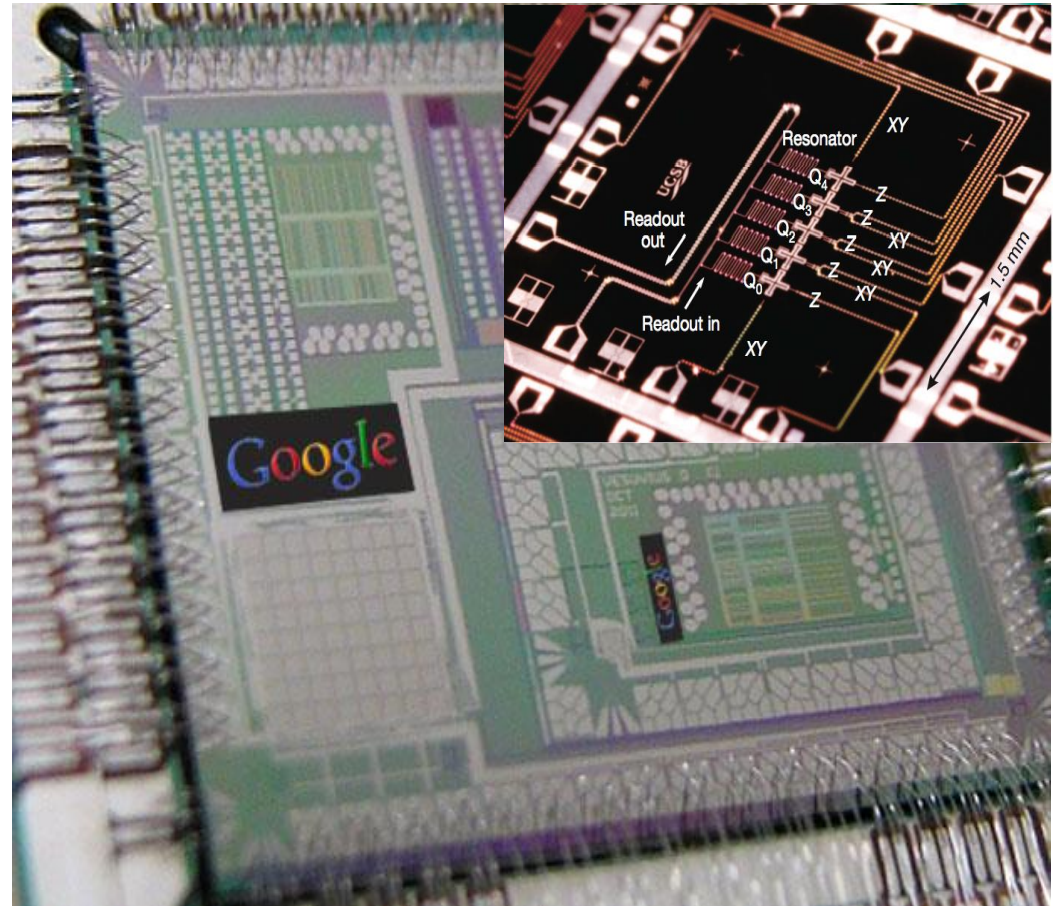
Martinis group, Nature **508**, 500 (2014)

谷歌新闻

2014年9月3日谷歌宣布建造基于超导电子学的超快量子芯片

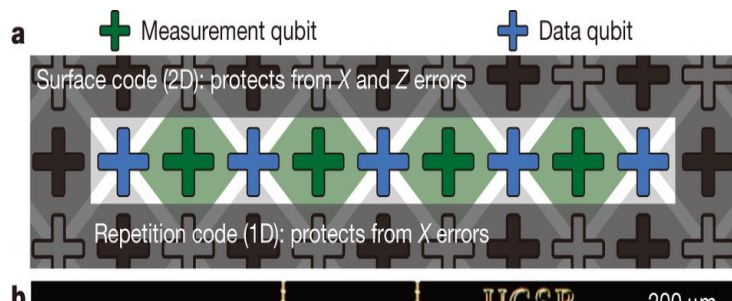


Google hires quantum computing expert John Martinis to build new hardware.

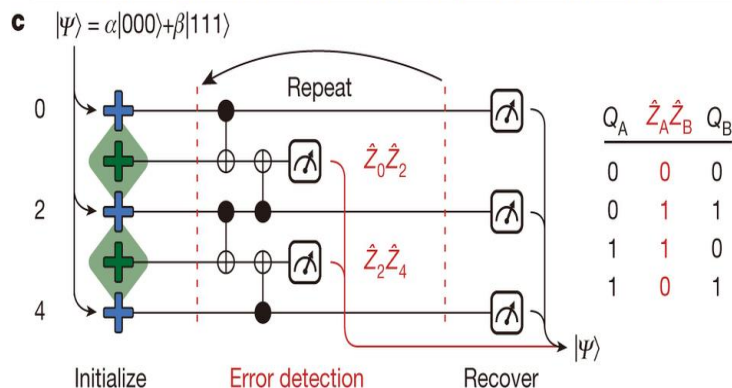
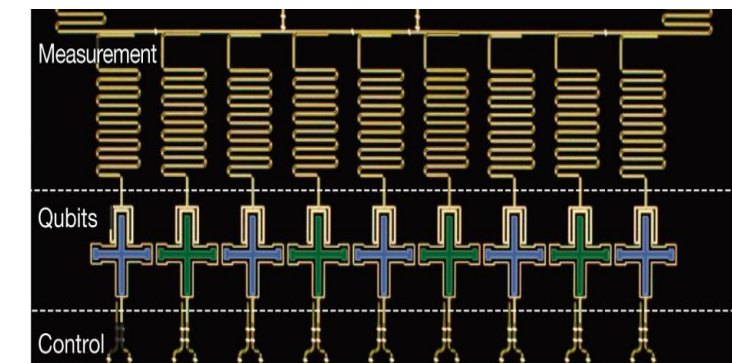


有关纠错和容错的实验

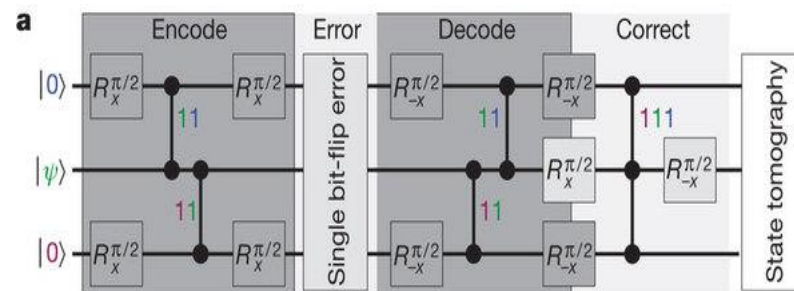
Xmon: 利用三比特编码的bit反转纠错



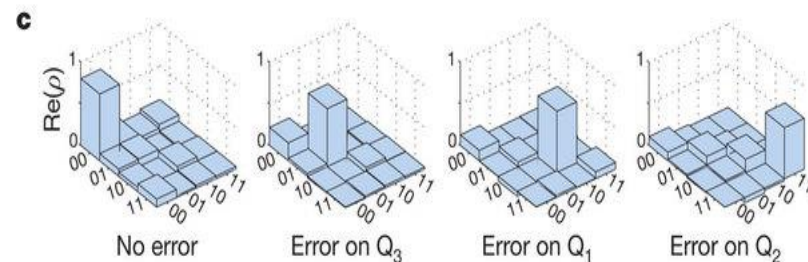
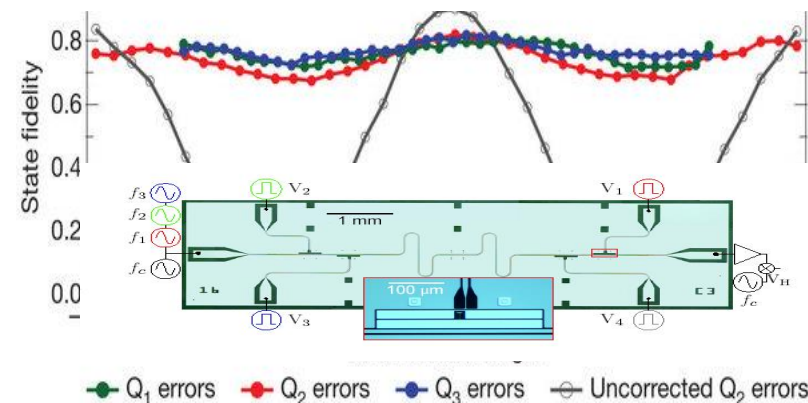
Martinis group: Nature **519**, 66(2015)



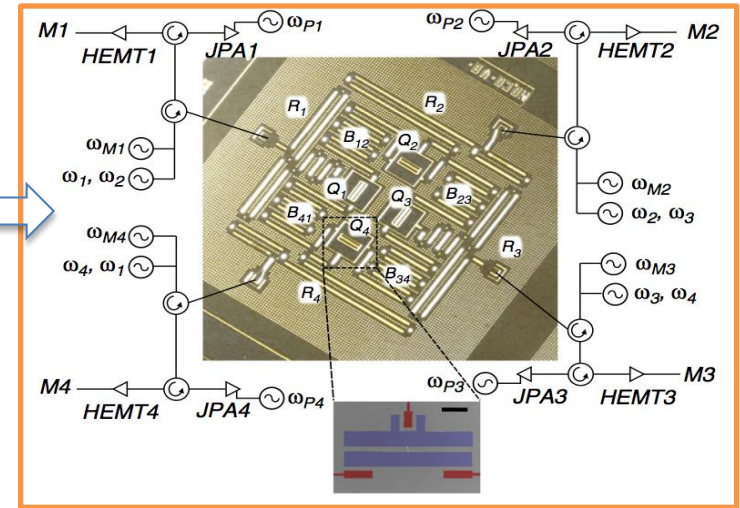
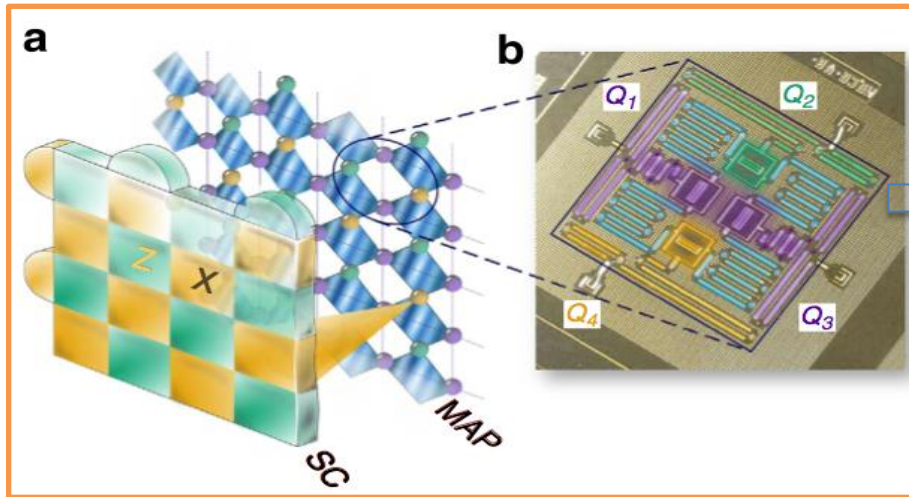
Transmon: 利用三比特编码的bit和phase 反转纠错



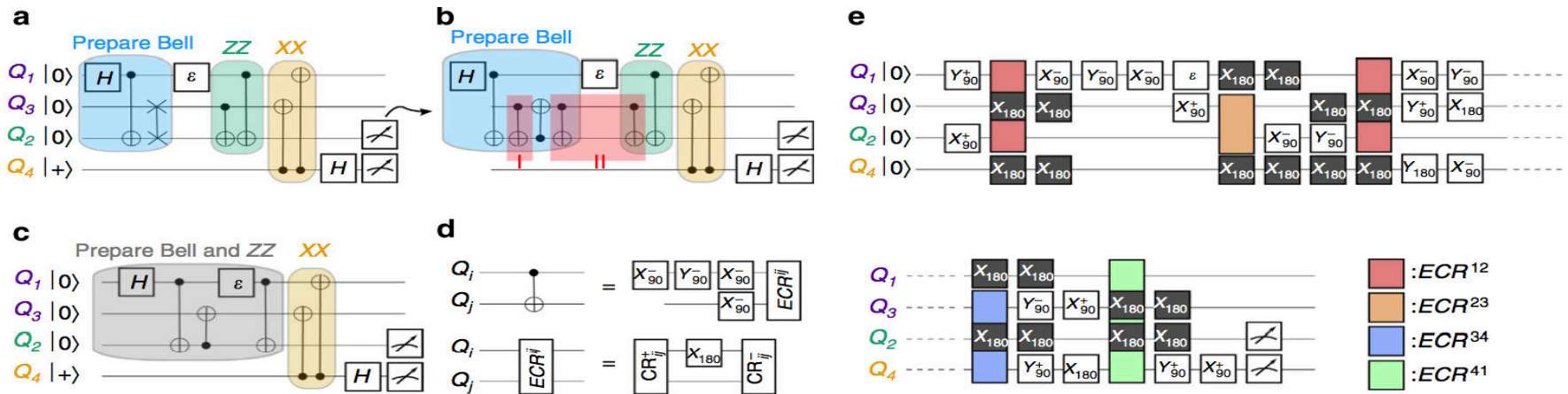
b Schoelkopf group, Nature **482**, 382 (2012)



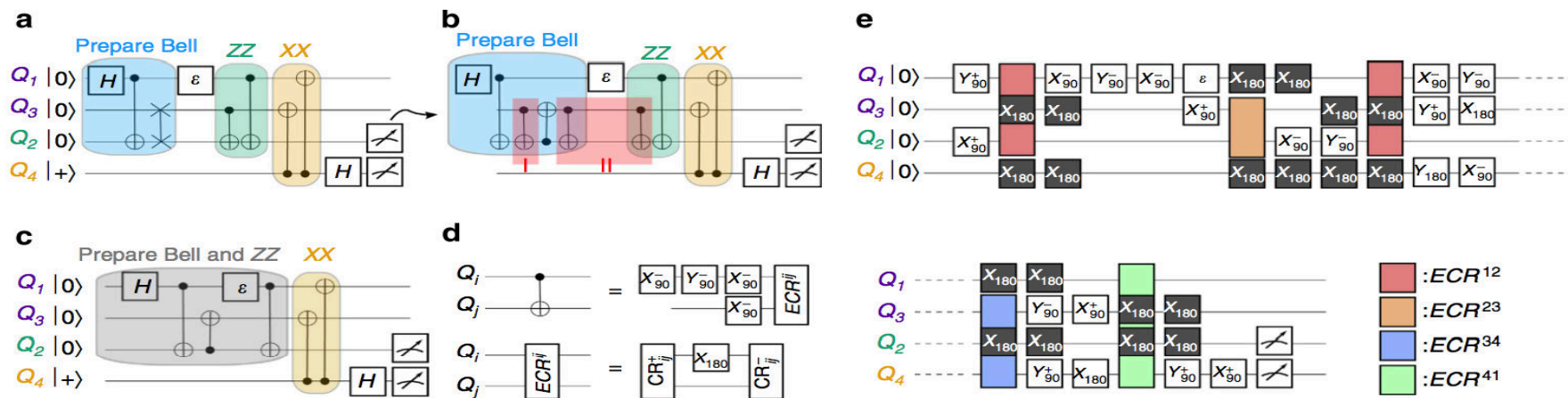
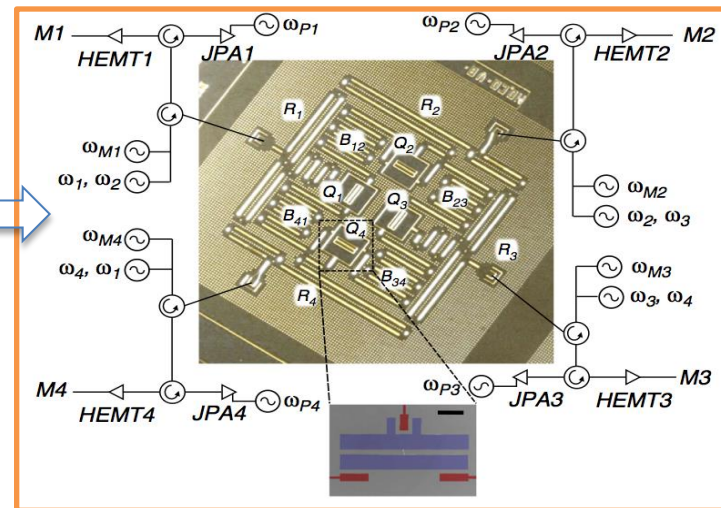
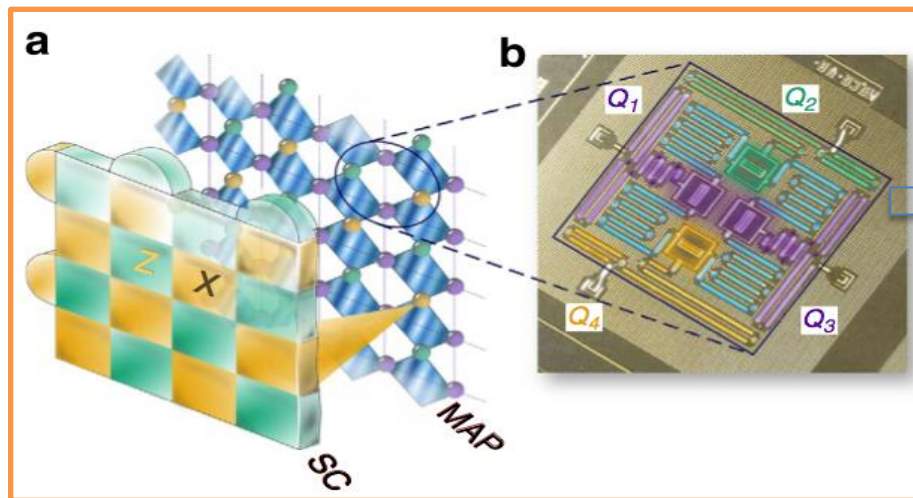
任意量子错误的探测实现量子纠错



主要问题：太多冗余



IBM组在Transmon 量子比特中实现了利用2比特编码的任意量子错误的探测和量子纠错

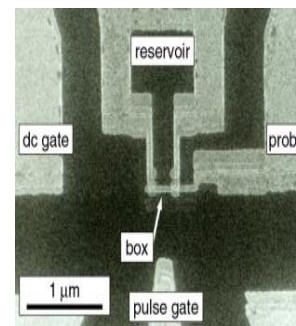
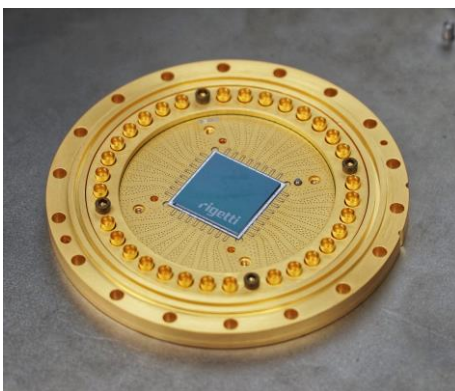
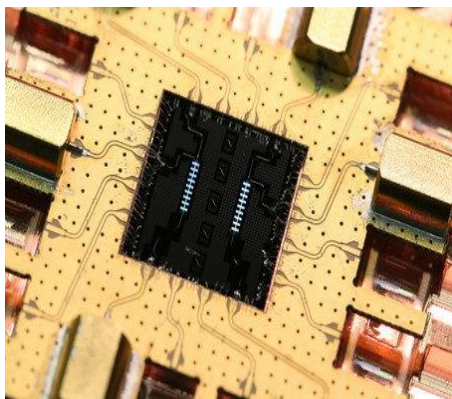


IBM group, Nature Commun. 5, 4015(2014); Nature Commun. 6, 6979 (2015)

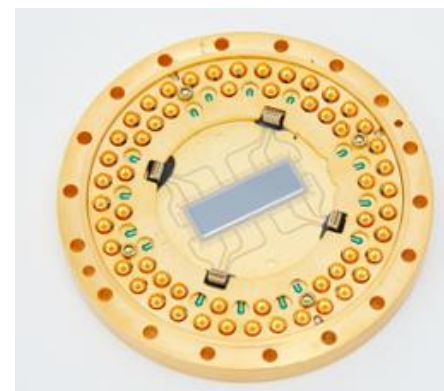
2014年后，量子计算机研发的公司雨后春笋般的出现，
Google, IBM, Intel, Microsoft, D-wave, Rigetti, Toshiba, IMEC, BBN Technologies.

国内：华为、腾讯.....。

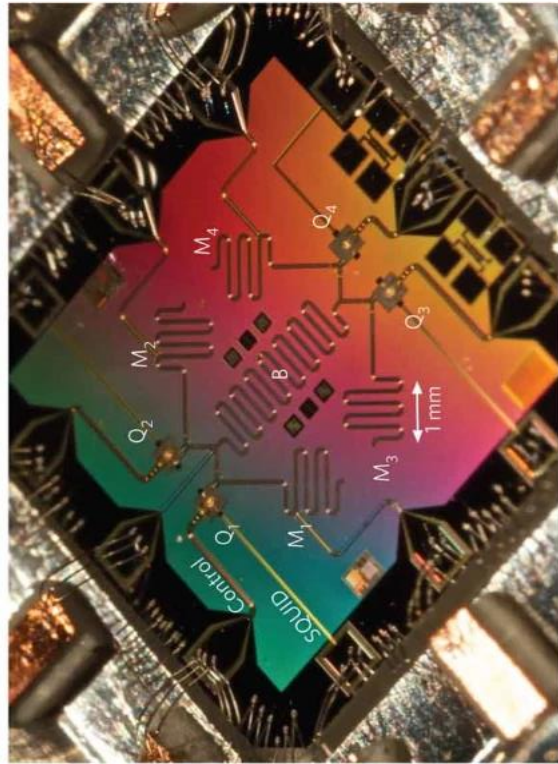
多量子比特的超导量子芯片



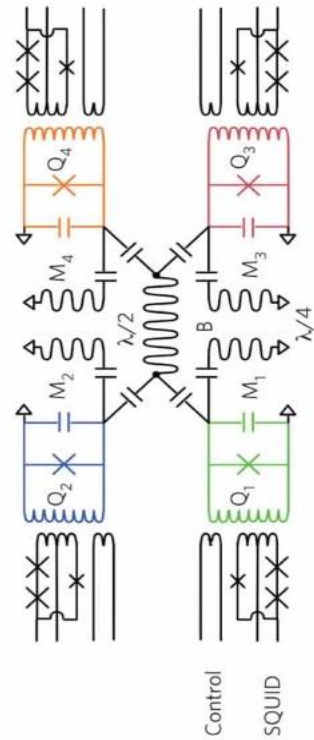
蔡兆申：1999年NEC



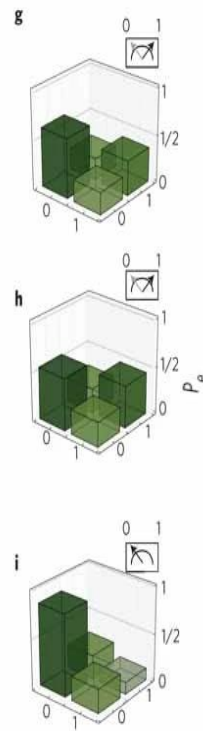
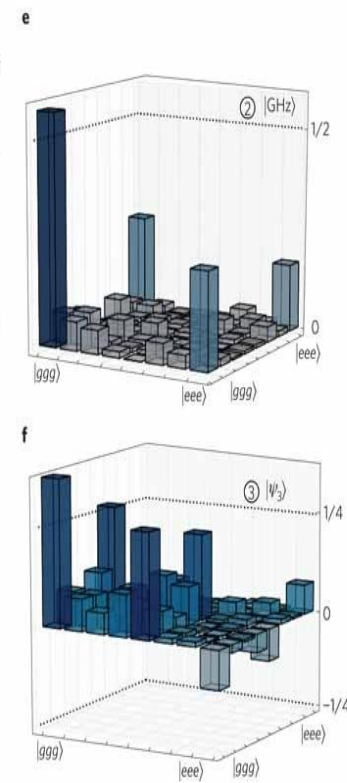
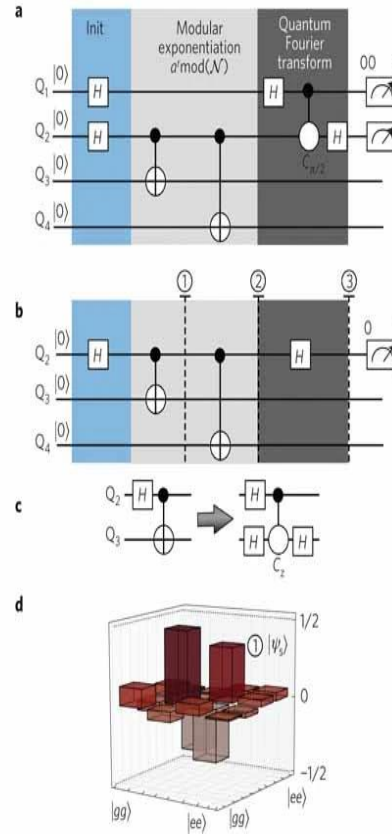
因子化的演示实验



a



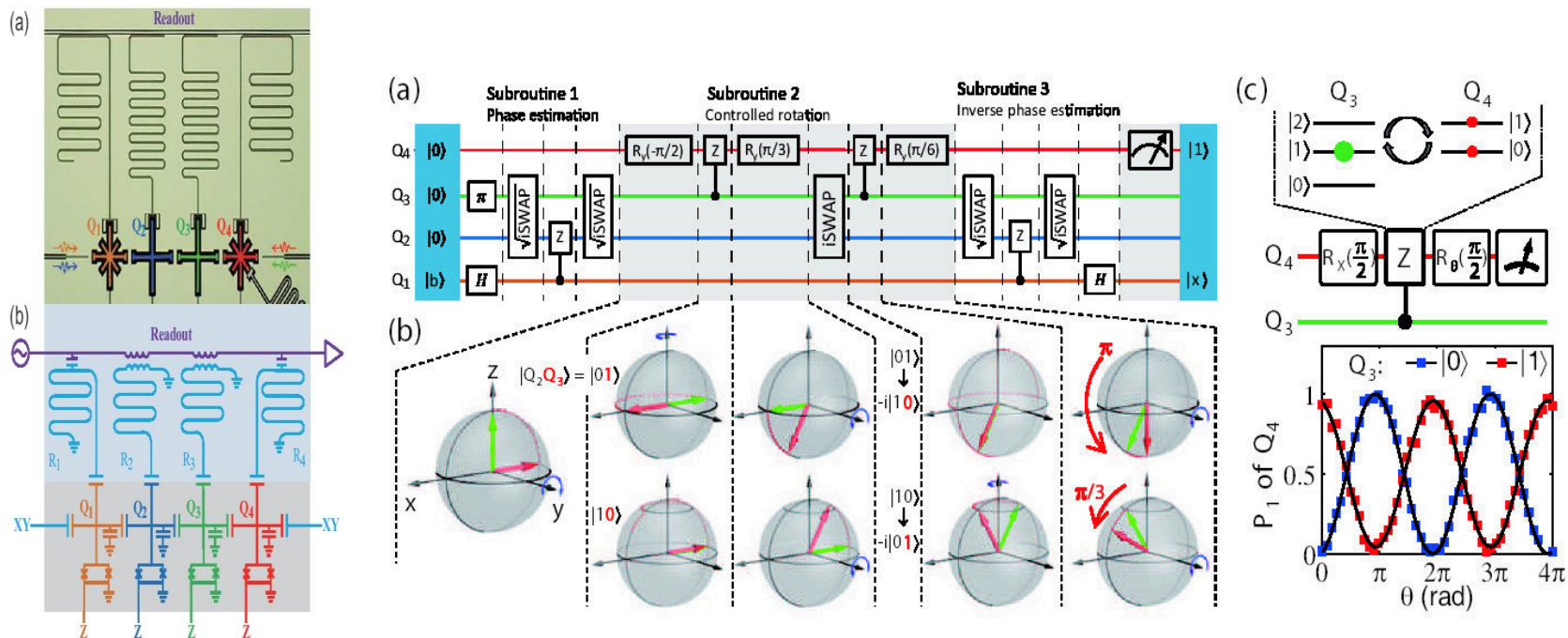
b



Nature Physics 8, pages 719–723 (2012)

解线性方程

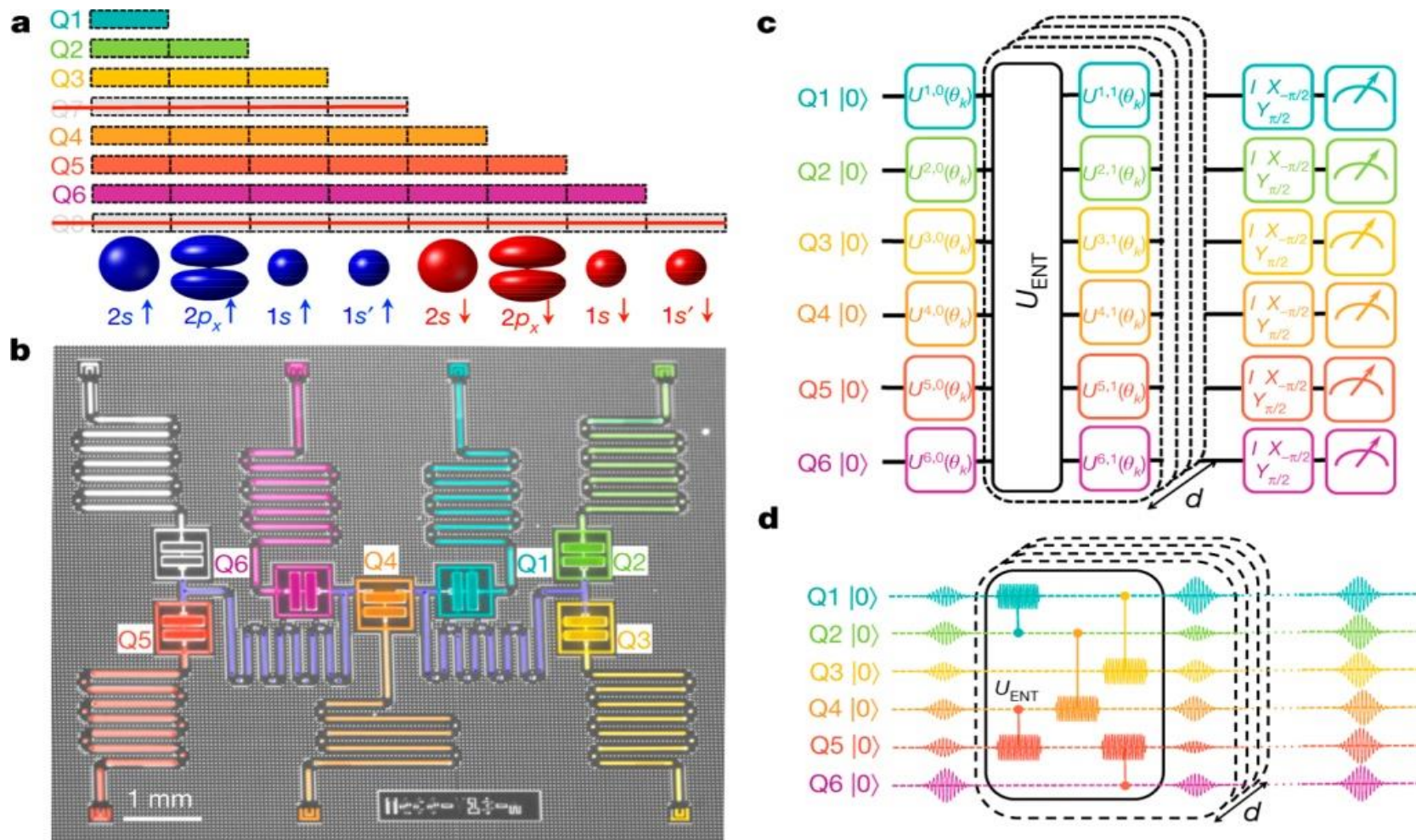
利用四个超导量子比特演示了可以解一个特殊的2X2的线性方程，并不是任意的2X2的线性方程，此特殊线性方程的系数矩阵为：



科大、浙大、物理所, Phys. Rev. Lett. 118, 210504 (2017)

结论：上述实验至多可以说演示了某种可能性，但是说可以解2X2的线性方程目前为时尚早，解任意N元一次方程的路更长。

分子基态的仿真

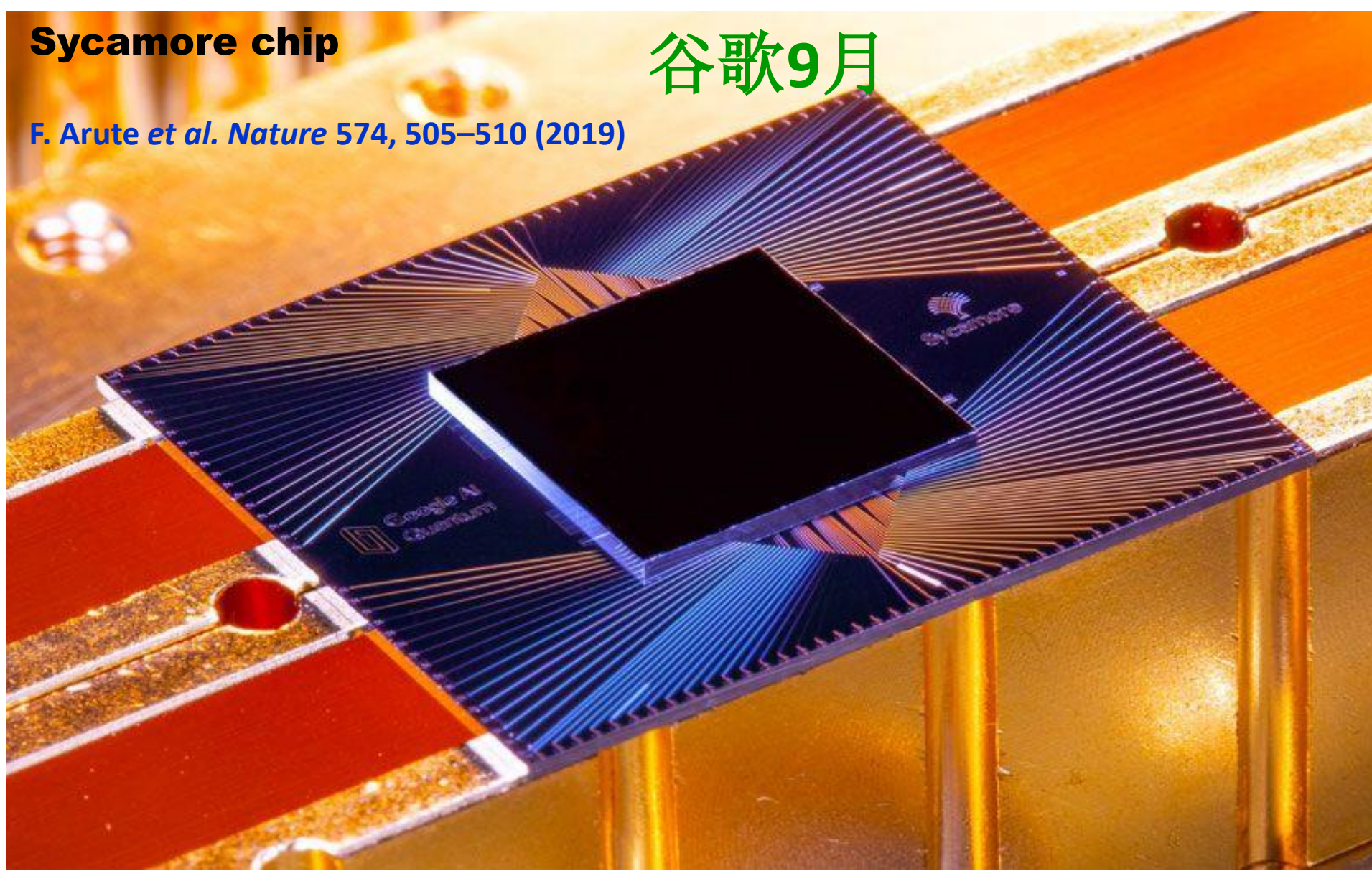


A Kandala *et al.* *Nature* **549**, 242–246 (2017), IBM组

Sycamore chip

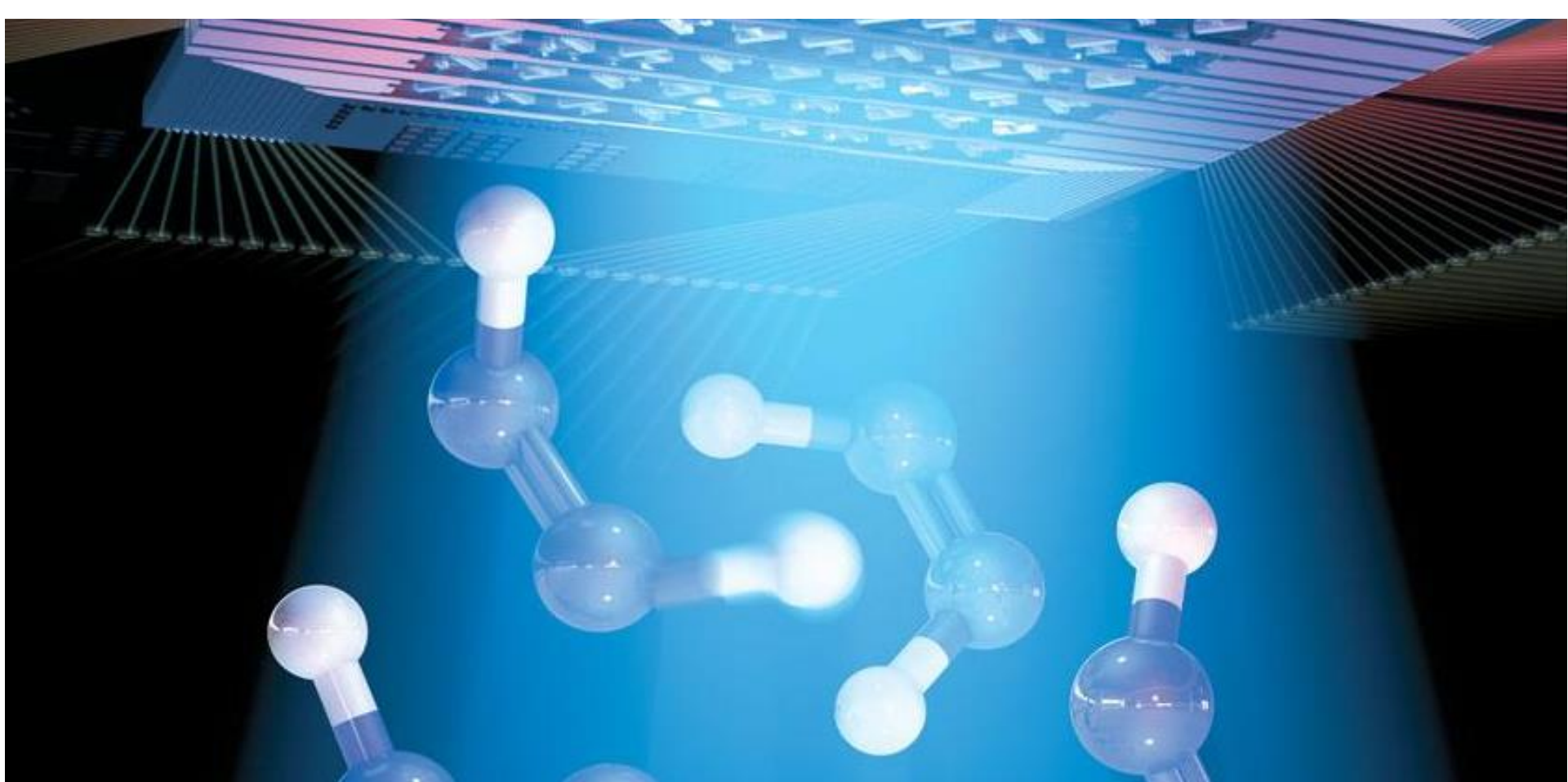
谷歌9月

F. Arute et al. Nature 574, 505–510 (2019)

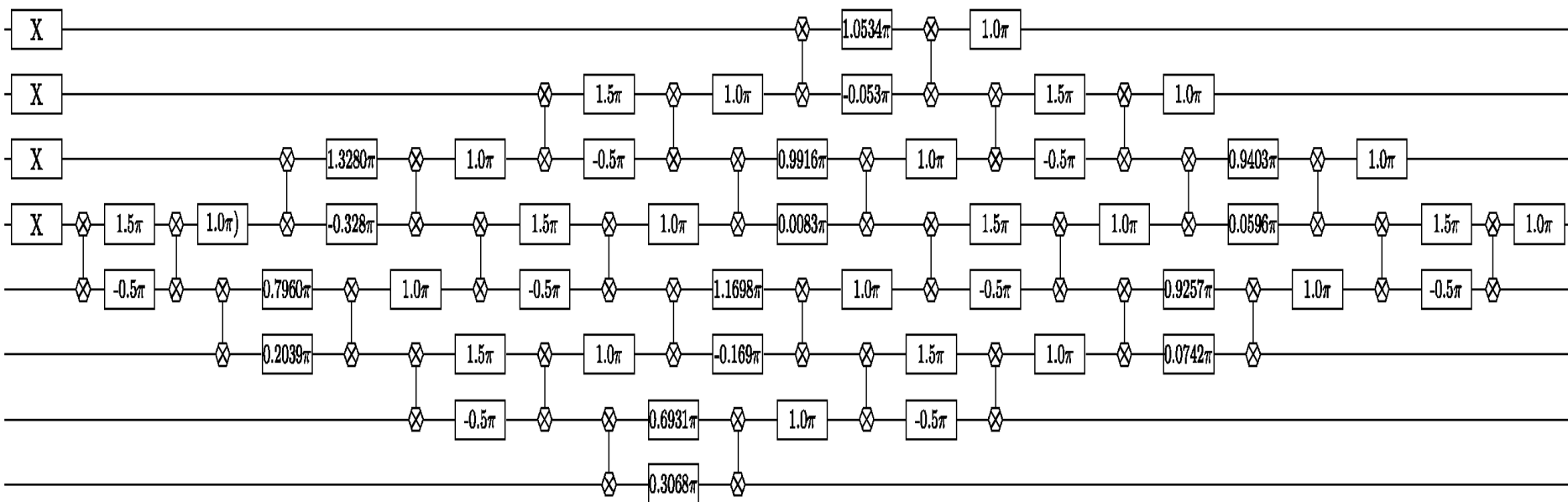


实际上执行运算的只有 53 个量子比特和 86 个耦合器，剩下的 1 个量子比特无法正常工作。200秒内完成的目标计算，最快的超级计算机则需要一万年。

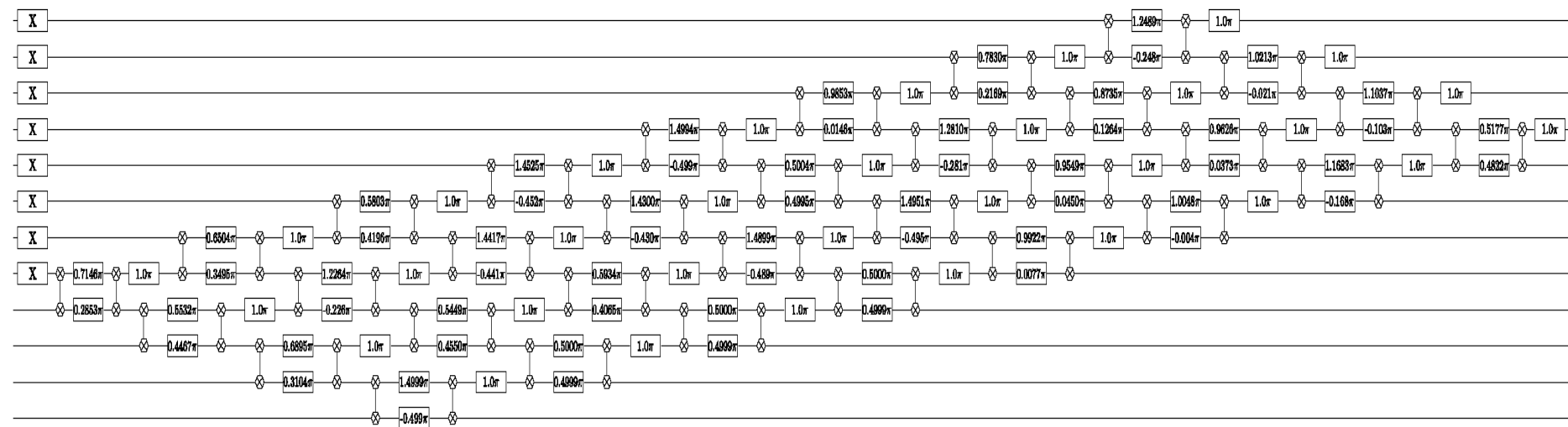
谷歌利用Sycamore量子处理器，完成了中等大小的化学问题的量子变分仿真。二亚胺（diazene）异构化机制和氢键结合能的仿真。



Google, Science **369**, 1084 (2020), 八月二十八

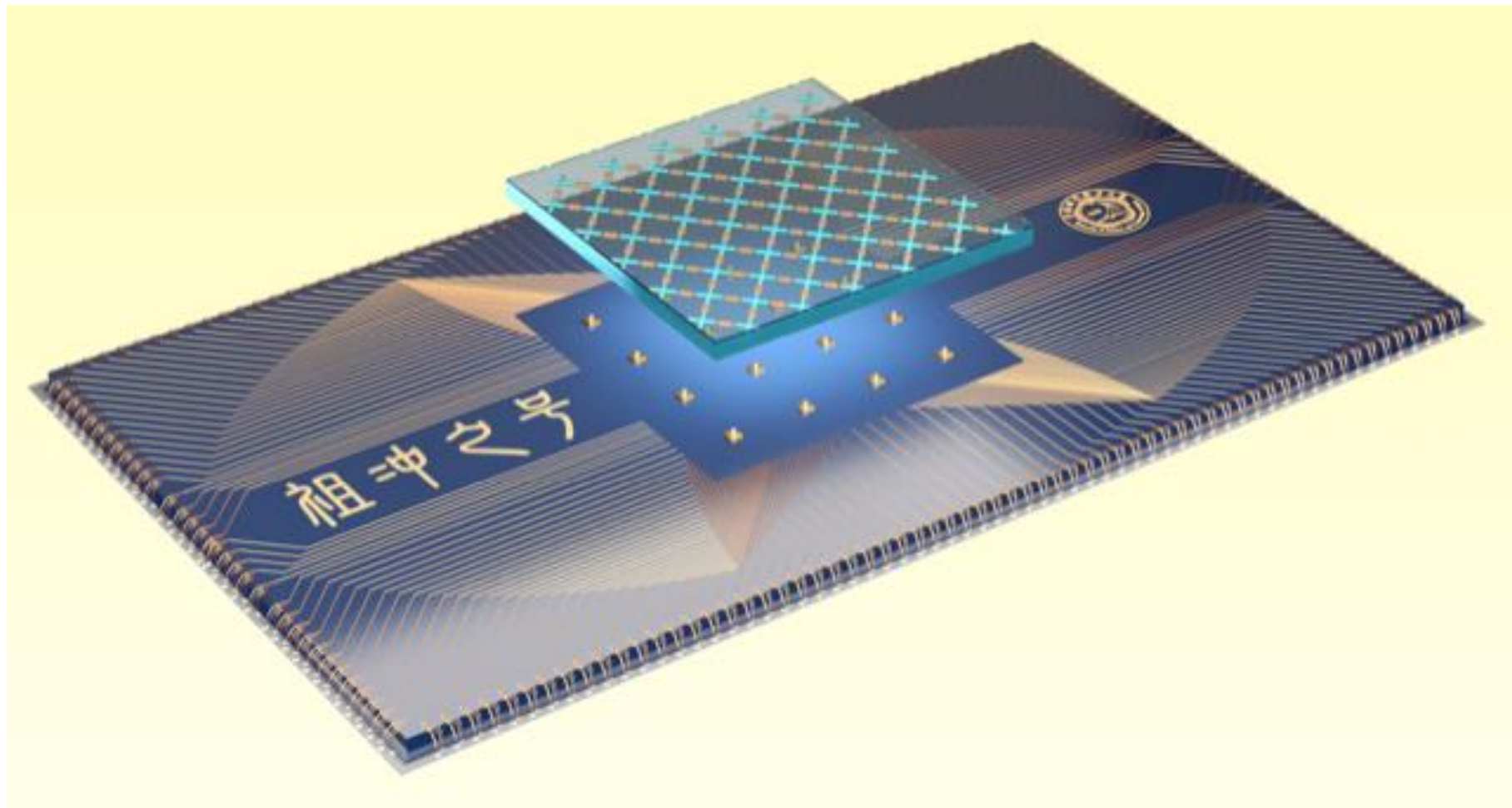


H₈链结合能线路图



二亚胺（ diazene ）异构化机制线路图

超导量子计算原型机 “祖冲之2.0”



是66比特可编程超导量子计算原型机“祖冲之2.0”的工作，研究人员通过操控其上的56个量子比特，在随机线路采样任务上实现了量子计算优越性。

- 1. Yale、UCSB、Chicago、Fermi;
 - 2. Intel 、IBM、 Microsoft、 **IMEC**
 - 3. Google、 BBN Technologies、 Rigetti, D-Wave
- 雷神子公司、量子技术

Spin qubits, Superconducting qubits, Cryo 3D integration, Cryo CMOS device testing, and modeling

Article

Advanced CMOS manufacturing of superconducting qubits on 300 mm wafers

<https://doi.org/10.1038/s41586-024-07941-9>

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J. Van Damme^{1,2}, S. Massar¹, R. Acharya¹, Ts. Ivanov¹, D. Perez Lozano¹, Y. Canvel¹, M. Demarets^{1,2}, D. Vangoldenhoven¹, Y. Hermans¹, J. G. Lai¹, A. M. Vadiraj¹, M. Mongillo¹, D. Wan¹, J. De Boeck^{1,2}, A. Potočník^{1,3} & K. De Greve^{1,2}

比利时: **IMEC**

¹Imec, Leuven, Belgium. ²Department of Electrical Engineering (ESAT), KU Leuven, Leuven, Belgium. [✉]e-mail: anton.potocnik@imec.be



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PAPER

Development of transmon qubits solely from optical lithography on 300 mm wafers

N Foroozani^{1,2}, C Hobbs³, C C Hung^{1,2}, S Olson³, D Ashworth³, E Holland³, M Malloy³, P Kearney³, B O’Brien³, B Bunday³, D DiPaola³, W Advocate³, T Murray³, P Hansen³, S Novak³, S Bennett³, M Rodgers³, B Baker-O’Neal³, B Sapp³, E Barth⁴, J Hedrick³, R Goldblatt³, S S Papa Rao³ and K D Osborn^{1,5,6}

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Multiplexed superconducting qubit control at millikelvin temperatures with a low-power cryo-CMOS multiplexer

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D. Perez Lozano¹, D. Wan¹, J. Van Damme^{1,2}, A. M. Vadiraj¹, M. Mongillo¹,
B. Govoreanu¹, J. Craninckx¹, I. P. Radu^{1,4}, K. De Greve^{1,2}, G. Gielen^{1,2},
F. Catthoor^{1,2} & A. Potočník¹

ARTICLES

<https://doi.org/10.1038/s41928-020-00528-y>

nature
electronics



A cryogenic CMOS chip for generating control signals for multiple qubits

S. J. Pauka^{1,2,7}, K. Das^{2,7}, R. Kalra², A. Moini^{1,2}, Y. Yang¹, M. Trainer², A. Bousquet², C. Cantaloube²,
N. Dick², G. C. Gardner^{3,4}, M. J. Manfra^{3,4,5,6} and D. J. Reilly^{1,2}✉

¹ARC Centre of Excellence for Engineered Quantum Systems, School of Physics, The University of Sydney, Sydney, New South Wales, Australia. ²Microsoft Quantum Sydney, The University of Sydney, Sydney, New South Wales, Australia. ³Birck Nanotechnology Center, Purdue University, West Lafayette, IN, USA. ⁴Microsoft Quantum Purdue, Purdue University, West Lafayette, IN, USA. ⁵Department of Physics and Astronomy, Purdue University, West Lafayette, IN, USA. ⁶School of Materials Engineering and School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN, USA. ⁷These authors contributed equally: S. J. Pauka, K. Das. ✉ e-mail: david.reilly@sydney.edu.au

<https://doi.org/10.1038/s41534-024-00840-x>

Systematic improvements in transmon qubit coherence enabled by niobium surface encapsulation

Mustafa Bal^{1,15}, Akshay Ziwon Huang¹, Tanay Andrei Lunin¹, Daniel B. Josh Y. Mutus³, Matthe Kameshwar Yadavalli³, Dominic P. Goronzy⁷, James M. Rondinelli⁷, Michael J. Bedzyk^{7,10}, Ma Jens Koch^{10,13}, James A. Sauls¹⁴, Alexander Ro

¹Superconducting Quantum Materials and Systems Division, Fermi National Accelerator Laboratory (FNAL), Batavia, IL 60510, USA. ²Department of Physics, Illinois Institute of Technology, Chicago, IL 60616, USA. ³Rigetti Computing, Berkeley, CA 94710, USA. ⁴Ames Laboratory, U.S. Department of Energy, Ames, IA 50011, USA. ⁵National Institute of Standards and Technology, Boulder, CO 80305, USA. ⁶Department of Electrical, Computer & Energy Engineering, University of Colorado, Boulder, CO 80309, USA. ⁷Department of Materials Science and Engineering, Northwestern University, Evanston, IL 60208, USA. ⁸The NUANCE Center, Northwestern University, Evanston, IL 60208, USA. ⁹International Institute of Nanotechnology, Northwestern University, Evanston, IL 60208, USA. ¹⁰Department of Physics and Astronomy, Northwestern University, Evanston, IL 60208, USA. ¹¹Department of Chemistry, Northwestern University, Evanston, IL 60208, USA. ¹²Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL 60208, USA. ¹³Center for Applied Physics and Superconducting Technologies, Northwestern University, Evanston, IL 60208, USA. ¹⁴Hearne Institute of Theoretical Physics, Department of Physics and Astronomy, Louisiana State University, Baton Rouge, LA 70803, USA. ¹⁵These authors contributed equally: Mustafa Bal, Akshay A. Murthy, Shaojiang Zhu, Francesco Crisa.

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PAPER

Building blocks of a flip-chip integrated superconducting quantum processor

Sandoko Kosen^{1,5,*}, Hang-Xi Li^{1,5}, Marcus Rommel¹, Daryoush Shiri¹, Christopher Warren¹, Leif Grönberg², Jaakko Salonen², Tahereh Abad¹, Janka Biznárová¹, Marco Caputo², Liangyu Chen¹, Kestutis Grigorasi², Göran Johansson¹, Anton Frisk Kockum¹, Christian Krizan¹, Daniel Pérez Lozano^{1,6}, Graham J Norris³, Amr Osman¹, Jorge Fernández-Pendás¹, Alberto Ronzani², Anita Fadavi Roudsari¹, Slawomir Simbierowicz^{2,7}, Giovanna Tancredi¹, Andreas Wallraff^{3,4}, Christopher Eichler³, Joonas Govenius² and Jonas Bylander¹

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³ ETH Zürich, CH-8093 Zürich, Switzerland

⁴ Quantum Center, ETH Zürich, CH-8093 Zürich, Switzerland

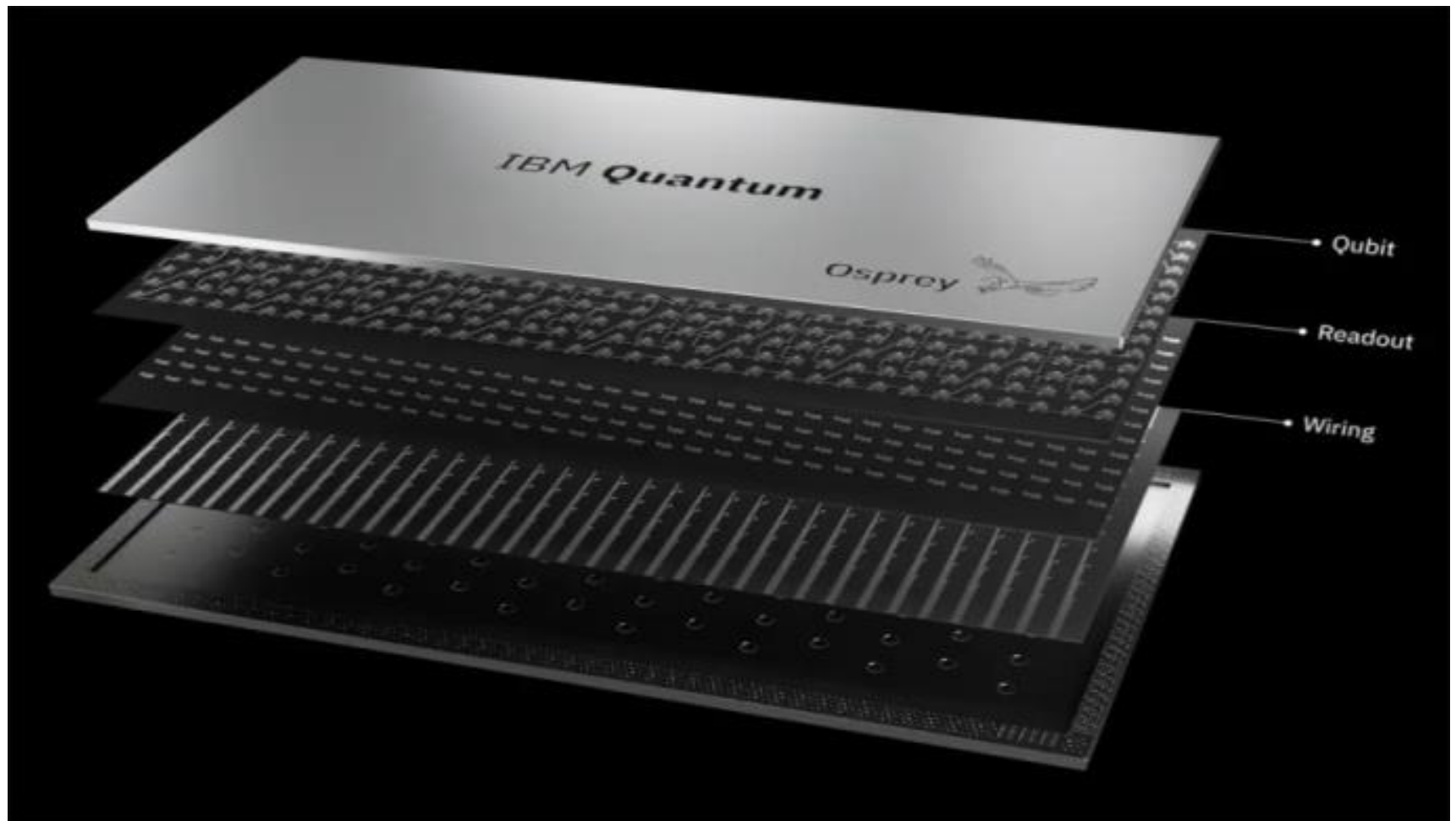
⁵ These authors contributed equally to this work.

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* Author to whom any correspondence should be addressed.

IBM超导量子计算发展路线图



IBM: 2021年; 127 qubits (Eagle); 2022年 433 qubits (Osprey); 希望2023年突破1000 qubits (Condor)

IBM超导量子计算发展路线图

Development Roadmap

Executed by IBM 
On target 

IBM Quantum

2019 2020 2021 2022 2023 2024 2025 Beyond 2026

Model
Developers

Prototype quantum software applications

Quantum software applications

Machine Learning | Optimization | Natural Science | Finance

Algorithm
Developers

Quantum algorithm and application modules

Quantum Serverless

Machine Learning | Natural science | Optimization | Finance

Intelligent orchestration

Circuit Knitting Toolbox

Circuit Libraries

Kernel
Developers

Circuits

Qiskit Runtime

Dynamic Circuits

Threaded Primitives

Error suppression and mitigation

Error correction

System
Modularity

Falcon
27 qubits

Hummingbird
65 qubits

Eagle
127 qubits

Osprey
433 qubits

Condor
1,121 qubits

Flamingo
1,386+ qubits

Kookaburra
4,158+ qubits

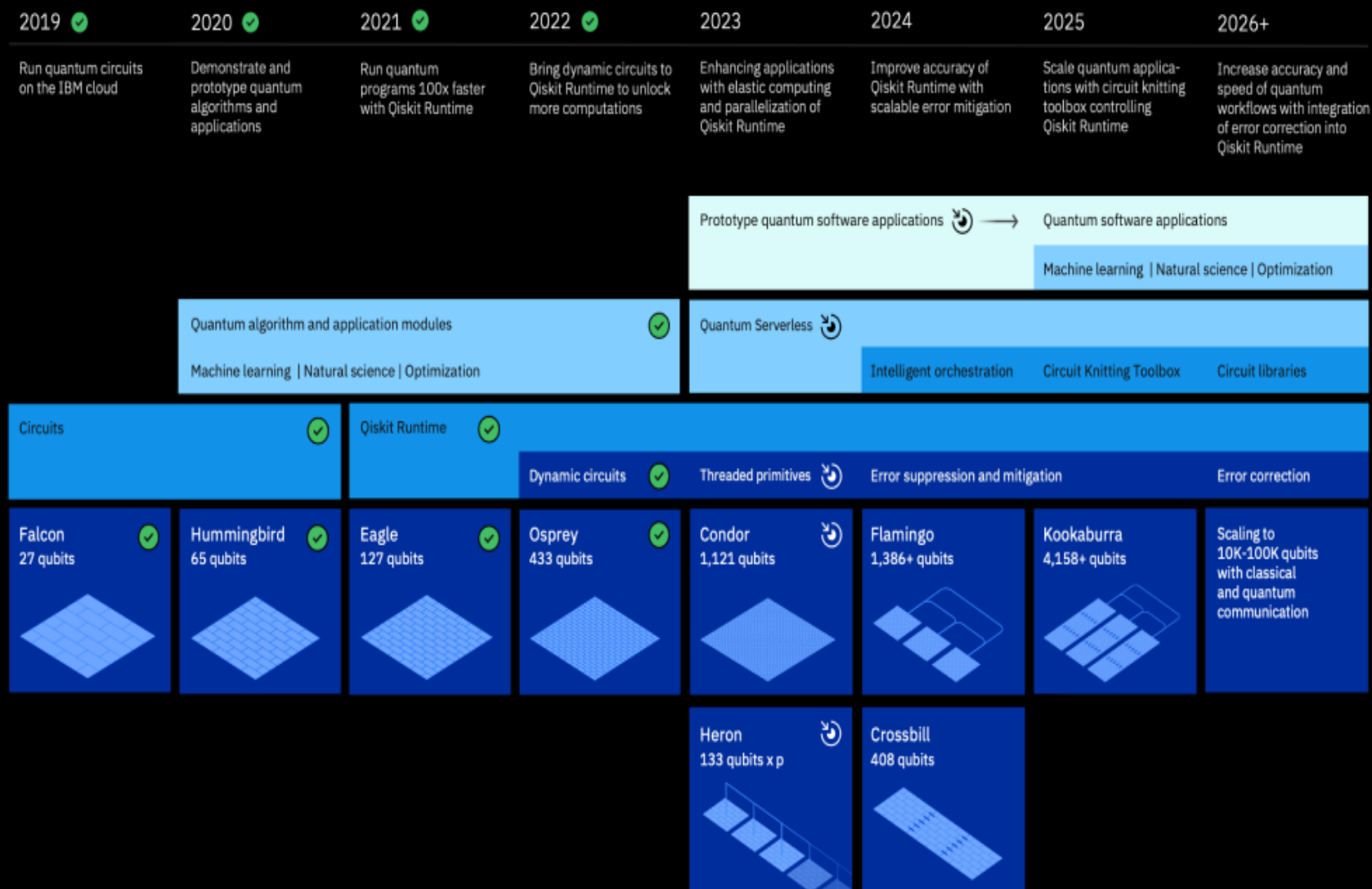
Heron
133 qubits x p

Crossbill
408 qubits

猎鹰、蜂鸟、鹰、鱼鹰、秃鹰、火烈鸟、笑翠鸟

2022年

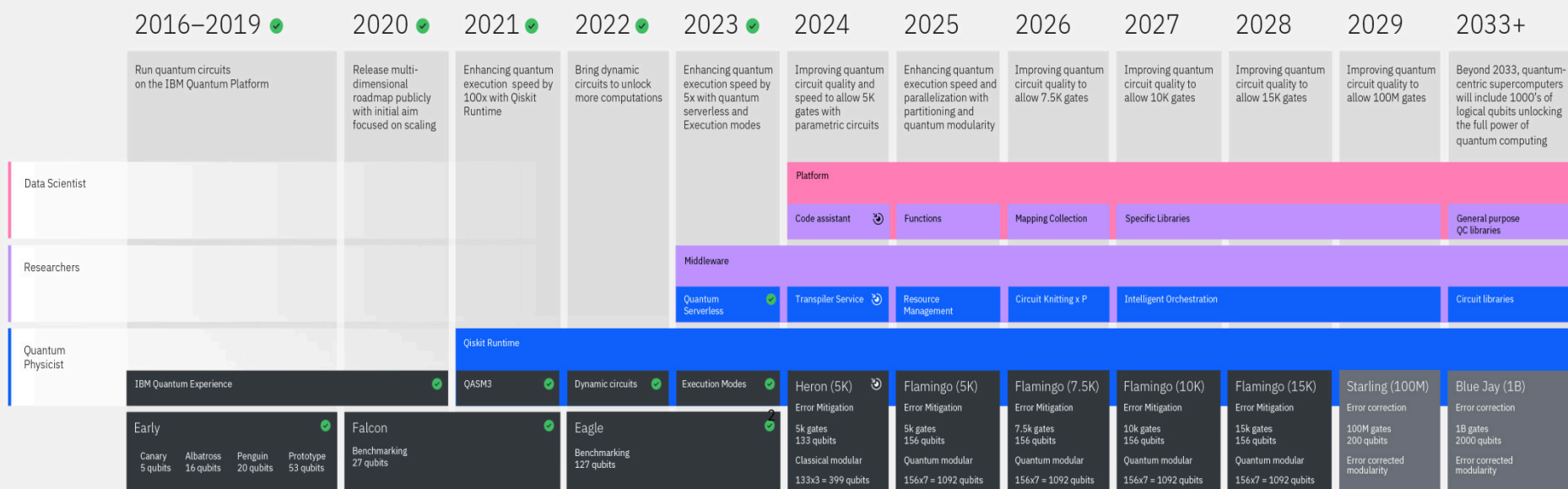
IBM超导量子计算发展路线图



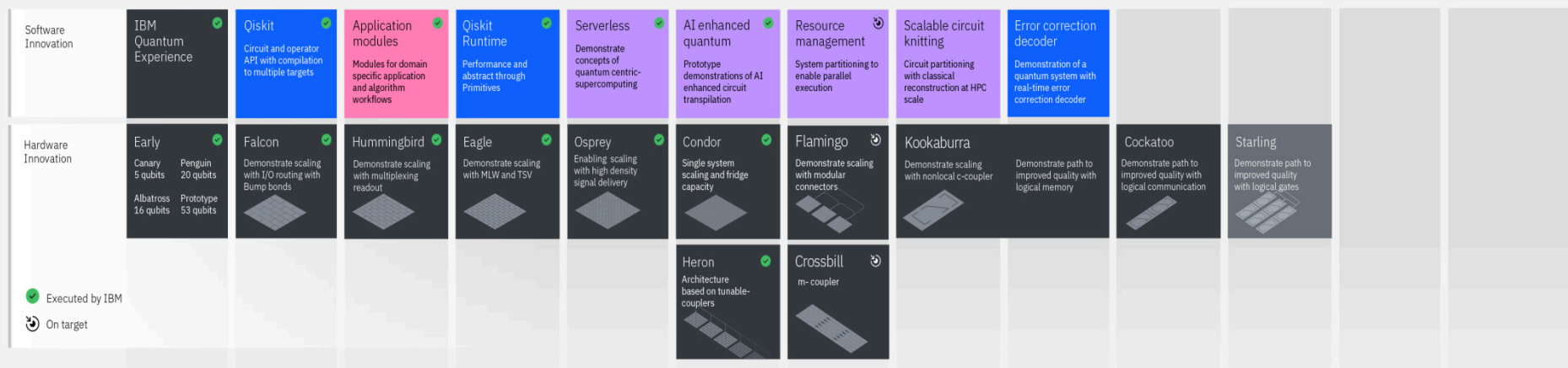
IBM超导量子计算发展路线图

Development Roadmap

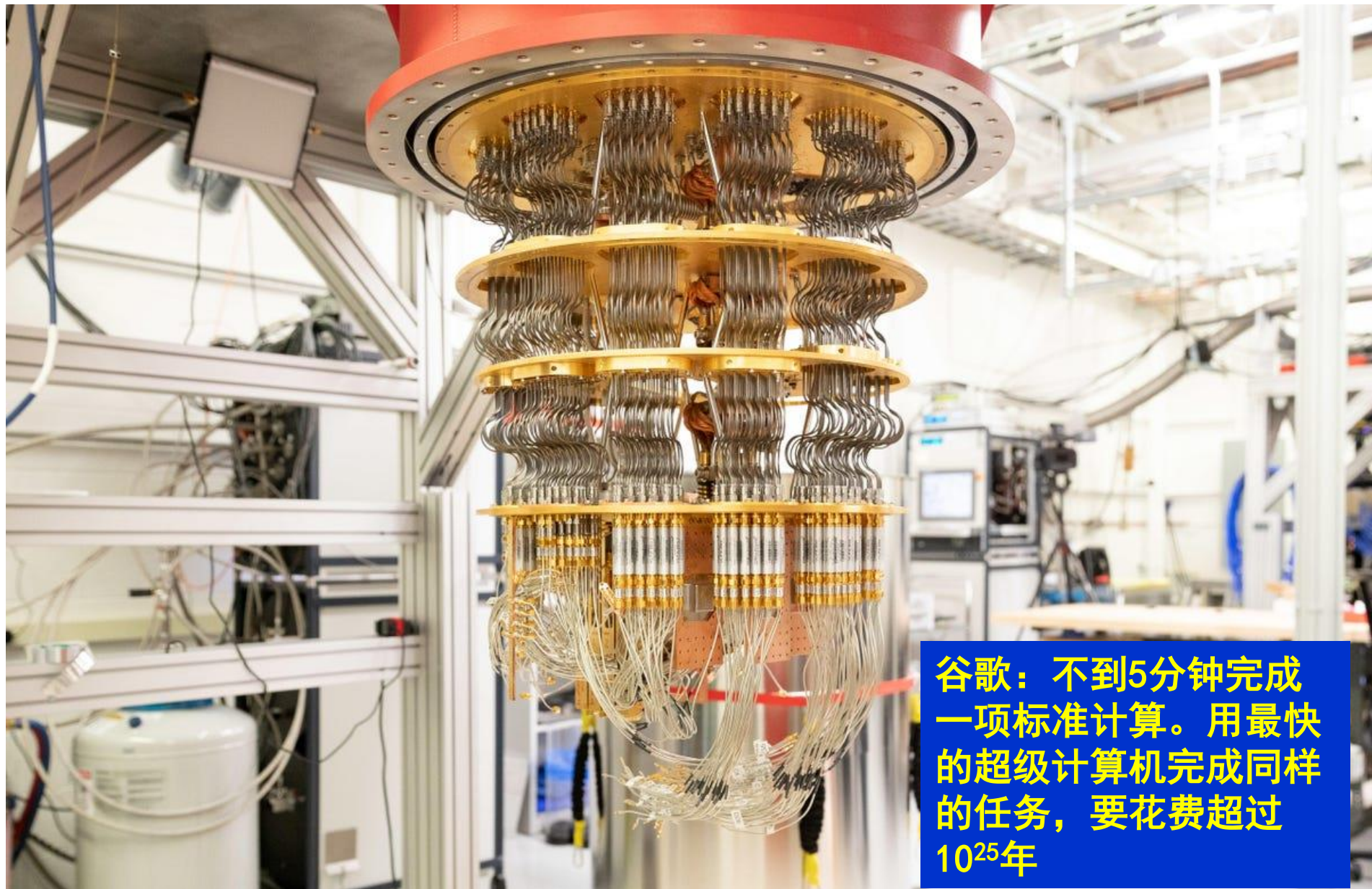
IBM Quantum



Innovation Roadmap



2024年12月10日：谷歌105个超导量子比特



谷歌：不到5分钟完成
一项标准计算。用最快的
超级计算机完成同样的
任务，要花费超过
 10^{25} 年

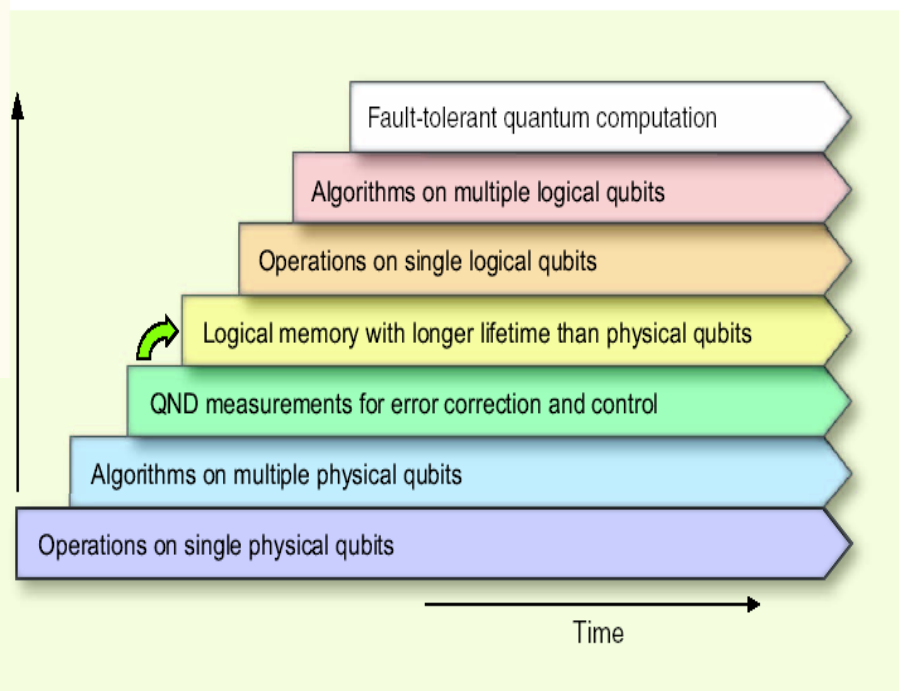
总之

量子计算超导**硬件**性能的发展取得了飞跃式的发展，量子信息处理最终需要通过**算法**和**编程**来实现。

量子算法和**量子程序**是什么？算法我们知道一些，如Shor因子化。量子编程，程序又怎么运行等等都需要结合硬件的发展研究，未来有很多很多的问题，不管有多困难，我们已在路上……

The concept of solving problems with the use of quantum algorithms, introduced in the early 1990s (1, 2), was welcomed as a revolutionary change in the theory of computational complexity, but the feat of actually building a quantum computer was then thought to be impossible. The invention of quantum error correction (QEC) (3–6) introduced hope that a quantum computer might one day be built, most likely by future generations of physicists and engineers. However, less than 20 years later, we have witnessed so many advances that successful quantum computations, and other applications of quan-

tum information processing (QIP) such as quantum simulation (7, 8) and long-distance quantum communication (9), appear reachable within our lifetime, even if many discoveries and technological innovations are still to be made.

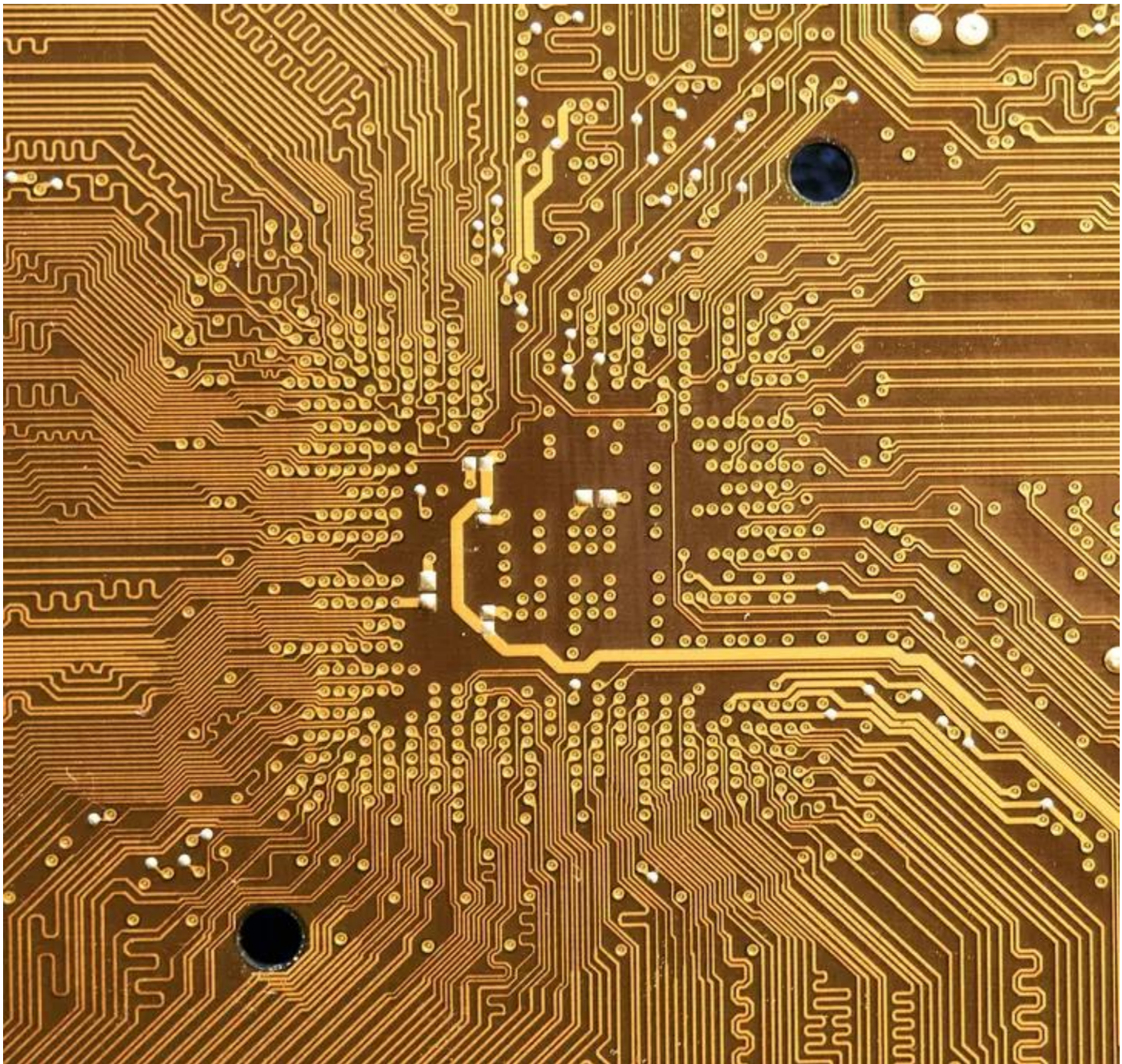


三部曲：

从不可能到可能再到有生之年看到

调研报告题目：

- 1、超导量子计算的进展
- 2、低温CMOS与量子计算
- 3、量子计算实现的硬件分类及优缺点



超导量子集成电路与量子计算

研究对象

基于约瑟夫森结的超导量子集成电路

内容

1. 量子逻辑的基本实现与控制
2. 面向应用的量子仿真与算法
3. 基于量子计算的人工智能

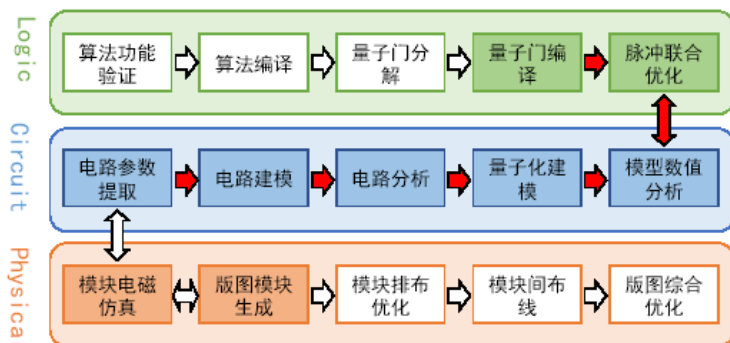
1. 量子计算器件的设计与仿真 (EDA)
2. 基本的物理性能
3. 集成超导电路的电子工程问题

1. Microwave resonator (微波谐振腔)
2. Microwave waveguide (微波波导)
3. Surface acoustic wave (表面声波)
4. Surface acoustic waveguide and resonator (表面声波导与谐振腔)
5. Mechanical resonator (机械谐振器)

广告: 我们的研究
欢迎合作, 欢迎做
博士后!!!

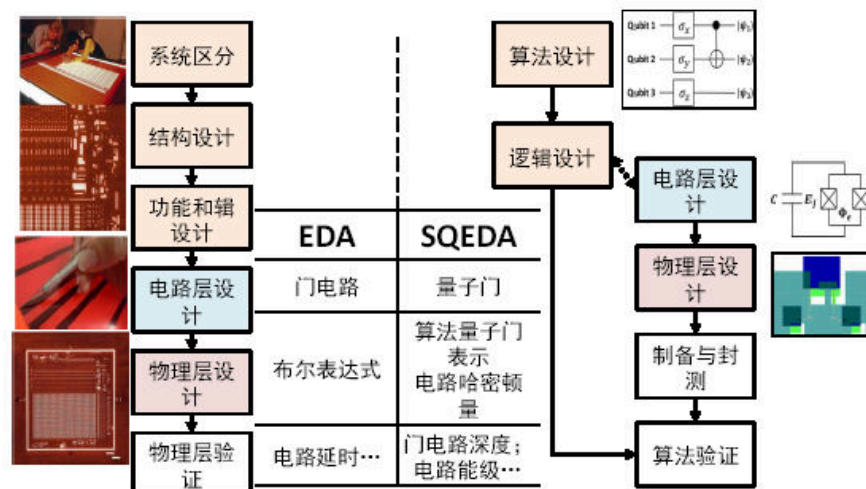
刘玉玺研究组

Roadmap for EDA of superconducting qubit circuits

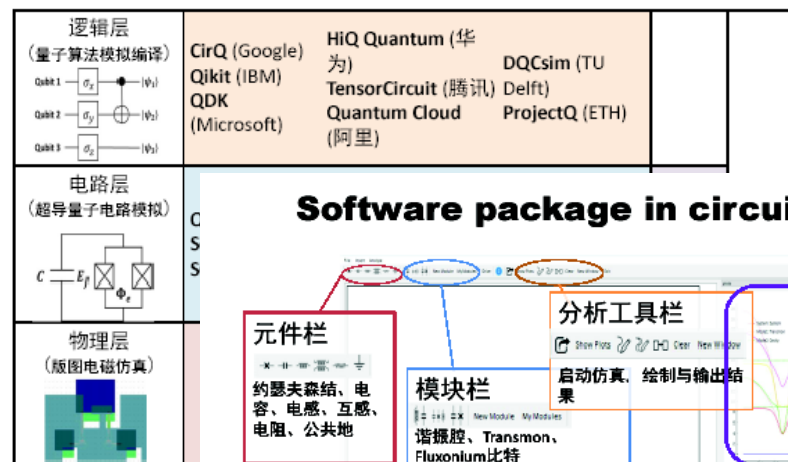


王佳恒同学领衔开发: 数值算法、微波工程, 机器学习、电磁仿真、量子算法等等

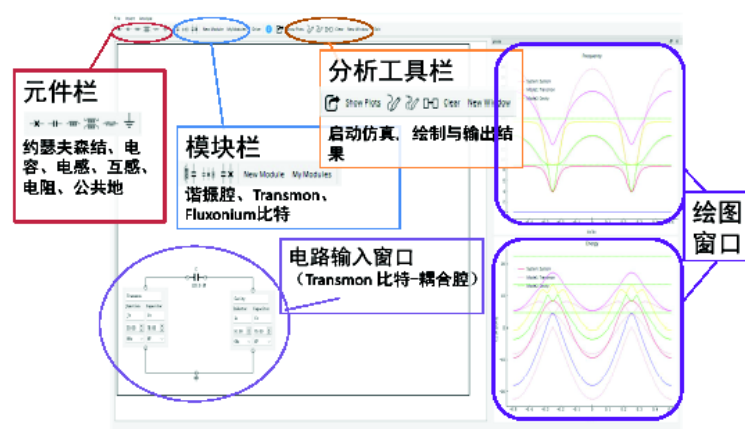
Electronic design automation for superconducting qubits



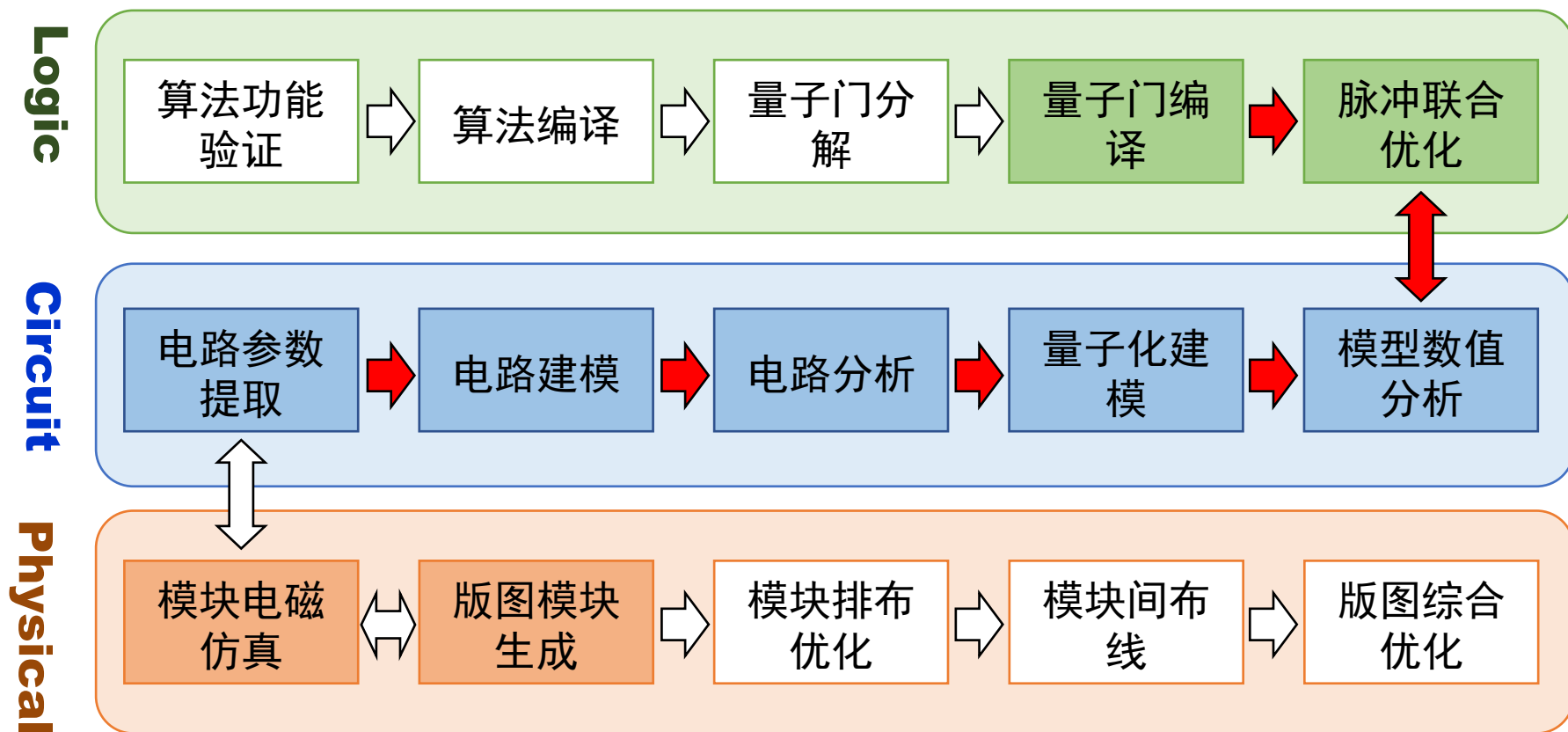
Open sources for EDA of superconducting qubit circuits



Software package in circuit layer

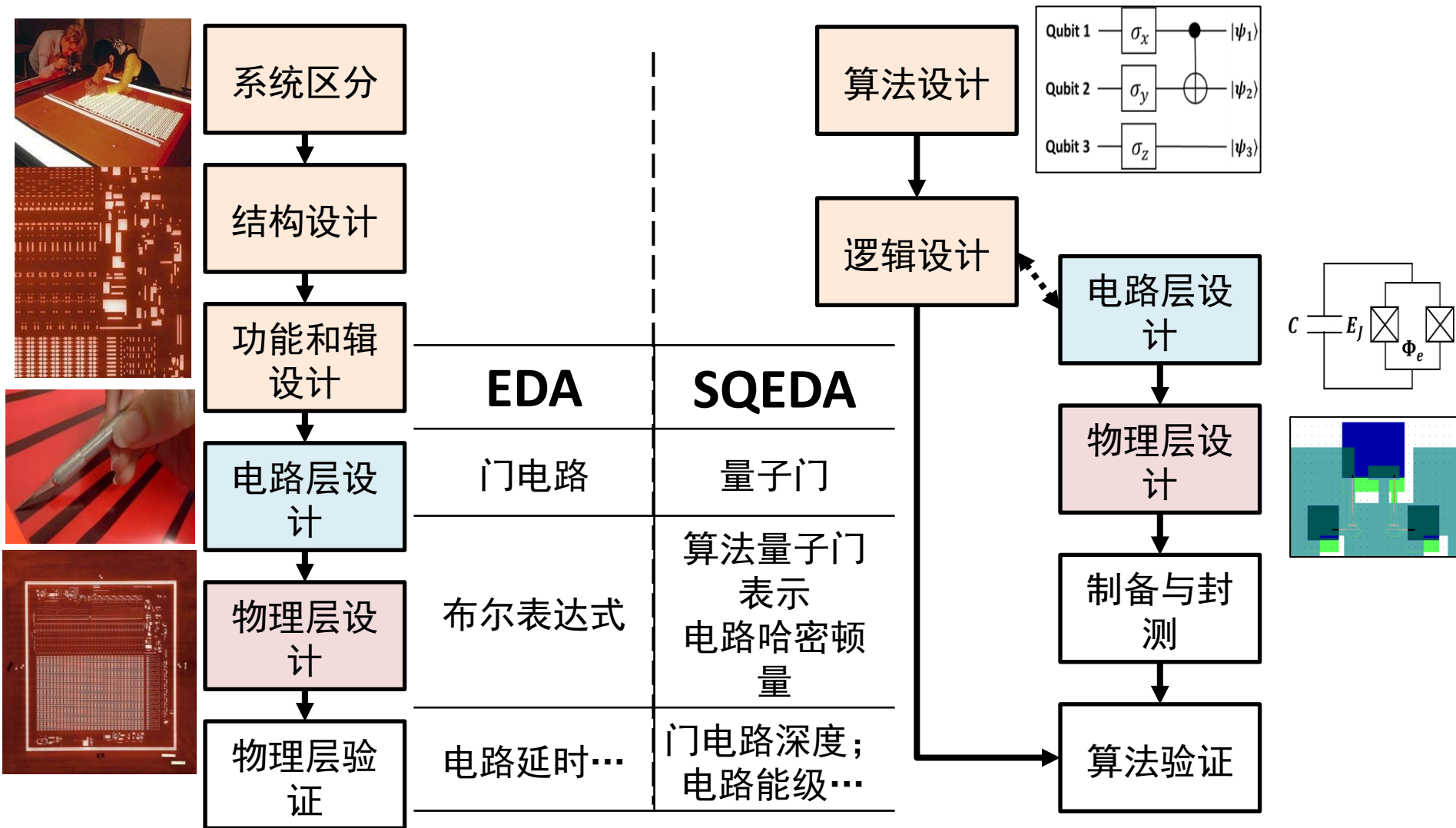


超导量子计算发展需要的所有模块

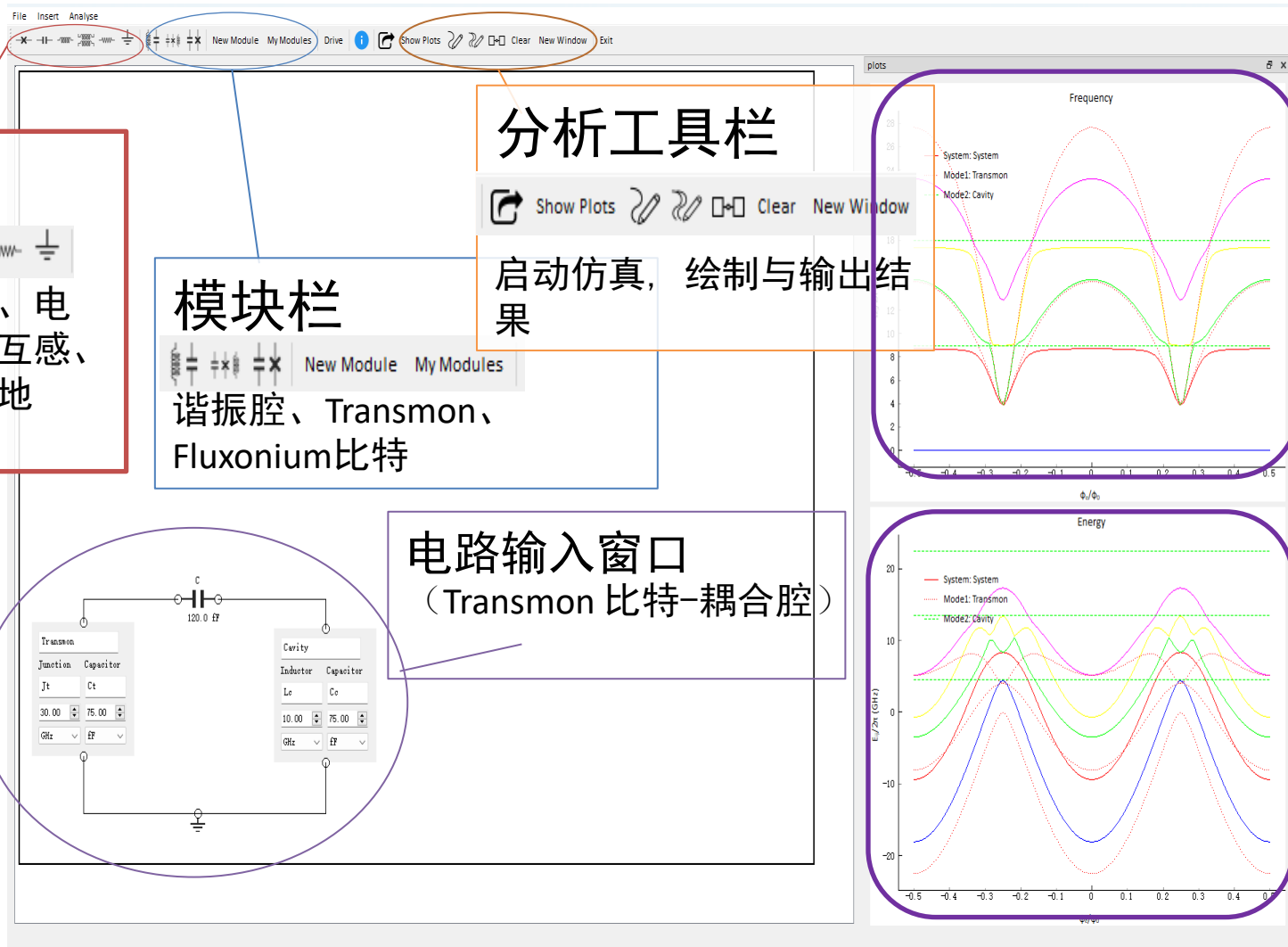


广告植入: yuxiliu@tsinghua.edu.cn

大规模超导量子芯片需要计算机辅助

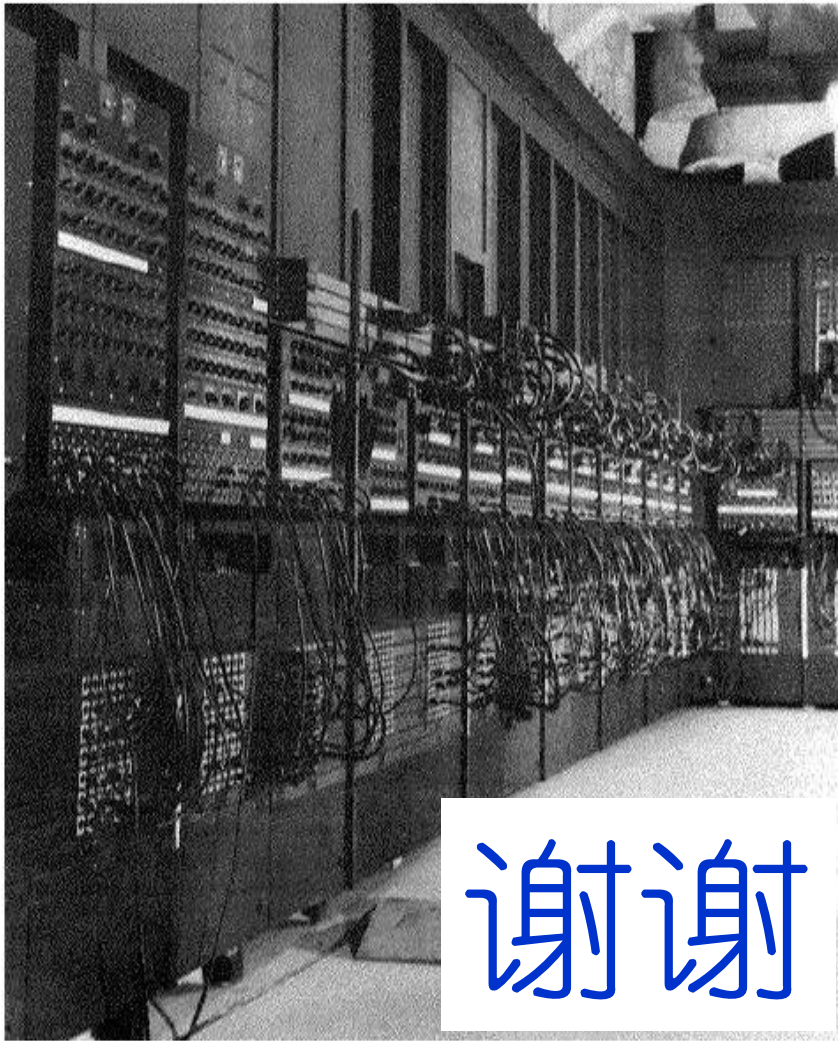


超导量子芯片设计仿真软件（电路层）



王佳恒同学等开发：数值算法、微波工程，机器学习、电磁仿真、量子算法等等

幻想还是现实？



20th century

about the ENIAC （电子数字积分计算机）

“Where a calculator on the ENIAC is equipped with 18000 vacuum tubes and weighs 30 tons, computers in the future may have only 1000 tubes and weigh only 1.5 tons.”

Popular Mechanics, March 1949